School of Innovation, Design and Engineering

A Framework for Embodied Carbon Reduction through Systematic Material Substitution in Modular Building Systems

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ABSTRACT

The construction sector is a major contributor to global greenhouse gas emissions, with material production representing an increasingly important share of total climate impact as buildings become more energy efficient. In modular building systems, reducing embodied carbon is complex because material choices are constrained by technical requirements, standardized interfaces, supplier conditions, implementation feasibility and circularity considerations. Existing research has largely focused on quantifying embodied carbon at building or system level, while providing limited guidance on how such results can be translated into structured material substitution decisions within existing modular product systems.

The purpose of this study was to develop and apply a structured approach for identifying and prioritizing high-impact materials within an existing modular building system, and to evaluate how potential material alternatives can be assessed while considering embodied carbon, technical feasibility and circularity. The study was conducted as a single case study of a modular classroom system. An abductive mixed-method design was applied, combining quantitative embodied carbon assessment with qualitative interview data. The quantitative analysis was limited to life cycle modules A1–A3 and was based on company-provided climate declaration data, material quantities, emission factors and product-specific Environmental Product Declarations. A Pareto-based hotspot analysis was used to identify dominant contributors to embodied carbon. The qualitative analysis was based on semi-structured interviews and a written questionnaire with company representatives, analyzed through thematic analysis.

The results show that embodied carbon within the studied module is concentrated in a limited number of materials. Facade board made of compressed insulation, fibre cement board and glass wool insulation 33 in the external wall were identified as the three dominant hotspot materials, together accounting for approximately 40 percent of total embodied carbon. The empirical findings show that material substitution decisions are shaped by system constraints, technical requirements, cost, supplier availability, implementation conditions and circularity ambitions.

Based on these findings, a structured six-step framework was developed, consisting of material mapping and system understanding, embodied carbon assessment, hotspot identification, substitution feasibility screening, multi-criteria evaluation, and implementation prioritization with iterative reassessment. The study concludes that embodied carbon reduction in modular building systems can be supported by combining hotspot-based prioritization with feasibility screening and multi-criteria evaluation. This enables companies to focus substitution efforts on high-impact materials while ensuring that technical, economic, organizational and circularity-related requirements are considered.

Keywords: embodied carbon, modular building systems, material substitution, hotspot analysis, multi-criteria decision-making, circularity, decision support, life cycle assessment

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- ABBREVIATIONS

CO_2	Carbon Dioxide
DfD	Design for Disassembly
EPD	Environmental Product Declaration
EN	European Standard
GHG	Greenhouse Gas
ISO	International Organization for Standardization
LCA	Life Cycle Assessment
MCDM	Multi-Criteria Decision-Making

1 INTRODUCTION

This chapter introduces the context and motivation for the study. It begins by outlining the global and national climate challenges associated with the buildings and construction sector, with particular emphasis on embodied carbon and material-related emissions. The chapter then narrows the focus to modular construction and discusses the complexity of material selection within industrial building systems. Based on previous research and identified practical challenges, the problem statement is formulated. Finally, the aim, research question, and delimitations of the study are presented.

1.1 Background

Climate change remains one of the most pressing global challenges of the 21st century, requiring substantial emission reductions across all sectors of society. Among these, the building and construction sector plays a particularly significant role. In 2023, the construction and operation of buildings accounted for approximately 34 per cent of global carbon dioxide (CO_2) emissions, making it the sector with the highest contribution worldwide (United Nations Environment Programme, 2025). These emissions arise from both operational energy use and the production of construction materials, underscoring the central importance of the sector in achieving global climate mitigation targets.

Although operational emissions have received significant attention over recent decades, embodied carbon has become increasingly important as buildings become more energy efficient. Embodied carbon refers to emissions associated with material extraction, manufacturing, transport, construction, maintenance and end-of-life processes. The Global Status Report emphasizes the need for adopting a whole life carbon perspective that integrates material-related emissions into decision-making processes throughout the building life cycle (United Nations Environment Programme, 2025). However, beyond immediate carbon intensity, material choices also influence durability, reuse potential and long-term resource efficiency. Research on circular construction highlights that climate mitigation must be aligned with circular economy principles, where materials are designed for longevity, disassembly and reintegration into future building systems (Ly et al., 2025). Consequently, embodied carbon assessments increasingly need to consider not only linear life-cycle emissions but also the retention of circular value over time.

In Sweden, the construction and real estate sector similarly constitutes a substantial share of national greenhouse gas (GHG) emissions. According to Boverket, the sector accounted for approximately 22.1 per cent of Sweden's total GHG emissions in 2022, corresponding to 10.8 million tons of CO_2 equivalents from domestic production (Boverket, 2025). When emissions from imported goods are included, total emissions amounted to 17.7 million tonnes of CO_2 equivalents. Life cycle assessments (LCAs) of the sector show that a significant share of indirect emissions originates from the production of construction materials (Boverket, 2025). These figures highlight not only the climate relevance of material-related emissions, but also the strategic importance of material selection in shaping future resource flows.

Modular construction has been promoted as a potential strategy for improving efficiency, reducing waste and enabling more controlled production processes. Prefabrication and industrialized construction methods may contribute to reduced construction time and improved material utilization. However, previous research demonstrates that modular construction does

not inherently guarantee lower climate impact. A critical review of the life cycle performance of modular buildings highlights substantial variation in reported environmental outcomes depending on material choices, system boundaries and modelling assumptions (Kamali & Hewage, 2016). Previous embodied carbon assessments indicate that environmental impacts are often concentrated to specific materials or life cycle stages, highlighting the importance of structured prioritization when analyzing complex building systems (Yan et al., 2025; Zhang et al., 2024).

From a circular perspective, modular construction can enable reuse and more controlled material flows throughout a building's life cycle, but the extent of that benefit depends on which materials are chosen and how they are integrated into the system. Early-stage design decisions are therefore critical, as they strongly influence a module's reuse potential, adaptability and long-term resource-efficiency performance (Dewagoda et al., 2024). Because material production accounts for a large share of embodied-carbon impacts in modular buildings, the selection of materials becomes central not only to immediate environmental footprints but also to the long-term circular performance of modular building systems (Zhang et al., 2024).

Life Cycle Assessment (LCA) is the standard methodological framework for quantifying environmental impacts across the building life cycle, including material extraction, construction, use and end-of-life stages (Ly et al., 2025). Recent LCA-based studies have used the method to identify environmental hotspots, repeatedly showing that material production often dominates the embodied-carbon profile of modular construction systems (Zhang et al., 2024). However, translating these hotspot results into systematic approaches for material prioritization within specific modular product systems remains difficult. Methodological variation, system complexity and context-specific constraints create a gap between LCA findings and practical decision-making in modular product development (Kamali & Hewage, 2016; Zhang et al., 2024).

1.2 Problem Statement

Despite the growing body of research on LCA and embodied-carbon quantification in construction, translating these environmental insights into actionable material decisions within industrialized modular building systems remains challenging (Ly et al., 2025; Zhang et al., 2024). Modular buildings consist of numerous standardized and interdependent components, meaning that material changes may affect structural performance, regulatory compliance and production processes (Dewagoda et al., 2024; United Nations Environment Programme, 2025). Existing studies mainly evaluate environmental performance at the building or system level, revealing environmental hotspots but providing limited guidance on how such results can be integrated into structured material decision-making processes during ongoing modular product development (Ly et al., 2025; Zhang et al., 2024). Consequently, companies must balance embodied-carbon reductions with technical requirements, cost constraints and circular-economy objectives without clear operational support for prioritizing which materials should be targeted first (Dewagoda et al., 2024; Ly et al., 2025). Previous studies show that a large share of embodied carbon in modular buildings is often associated with material production, highlighting the importance of prioritizing high-impact materials during early design and development stages (Zhang et al., 2024).

There is therefore a need for structured and operational approaches that support the systematic identification and prioritization of high-impact materials within existing modular systems,

while ensuring technical feasibility, regulatory compliance and consideration of circular performance requirements (Dewagoda et al., 2024; Ly et al., 2025).

1.3 Aim and Research questions

The purpose of this study is to develop a structured framework for identifying and prioritizing high-impact materials within an existing modular building system, and to design a multi-criteria approach for evaluating material alternatives that may reduce embodied carbon while considering technical feasibility and circularity.

1. *How can high-impact materials be systematically identified and prioritized within an existing modular building system to support reductions in embodied carbon?*

2. *How can a structured evaluation approach be designed to assess material alternatives for substitution in modular building systems, considering embodied carbon, technical feasibility, and circularity?*

1.4 Delimitations

This study focuses on embodied carbon associated with material production and construction stages within a selected modular building system, limited to life cycle modules A1–A3. Operational energy use during the building's use phase is not included within the system boundary. The embodied carbon assessment covers new materials only, as reused components, representing approximately 32 of the total 55 materials in the module, were excluded from the analysis since their production-stage emissions fall outside the defined system boundary. Environmental impact data is based on manufacturer-specific EPDs covering modules A1–A3. Transport from manufacturer to construction site (A4) and on-site construction processes (A5) are not included in the assessment.

The analysis focuses on material and component decisions within the context of a selected modular product system and does not include full building-level optimization across alternative architectural designs. The proposed approach is applied using available environmental data and company-provided material information in order to identify and prioritize high-impact materials and explore potential material alternatives. No experimental testing or physical validation of alternative materials is conducted, and structural feasibility is assessed based on documented technical requirements rather than engineering simulations or physical testing.

End-of-life aspects are considered qualitatively in relation to circular economy principles but are not modelled through detailed life cycle simulations. Furthermore, the findings are derived from a specific modular product system and are therefore not intended to provide universally generalizable results for all modular construction contexts.

2 LITTERATURE REVIEW/THEORETIC FRAMEWORK

2.1 Systems Thinking in Modular Product Architectures

2.1.1 Systems Thinking and Holism in Construction

Systems thinking describes an approach to understanding complex phenomena by focusing on relationships between elements rather than viewing parts in isolation. Arnold and Wade (2015) define systems thinking as a set of analytical skills used to identify, understand and predict system behavior and to guide purposeful modifications. A central idea in this perspective is that system outcomes depend on how components interact within a larger whole (Arnold & Wade, 2015; Godfrey, 2010).

In the built environment, sustainability can therefore be understood as an emergent property of complex systems. Godfrey (2010) explains that sustainability arises from the interaction of multiple components, meaning that overall performance cannot be explained by examining individual materials or technologies separately. Instead, sustainability results from how elements function together within their broader context (Godfrey, 2010). This contrasts with approaches that treat environmental improvements as isolated technical measures, a tendency identified in architectural sustainability research (Hensel, 2012).

Systems thinking also emphasizes the integration of multiple dimensions within a shared framework. Gonçalves & da Silva (2025) show that applying systems thinking in housing design enables the integration of technical, social, cultural and environmental considerations. From this perspective, components are best understood in relation to one another, and system outcomes emerge from their interactions (Gonçalves & da Silva, 2025).

2.1.2 Product Architecture: Modularity and Functional Allocation

Product architecture defines how a product's functions are allocated to its physical components (Ulrich, 1995). It describes how functional elements are arranged, how they are assigned to specific parts, and how these parts are connected through interfaces (Muffatto & Roveda, 2002; Ulrich, 1995). Because the architecture determines how functions and components relate to each other, it has a significant influence on product performance and manufacturing outcomes (Yin et al., 2014).

Product architectures are commonly described as either modular or integral (Ulrich, 1995; Veenstra et al., 2006). In a modular architecture, functions are largely assigned to separate components that are connected through standardized and decoupled interfaces (Ulrich, 1995). This structure limits the impact of changes, as modifying one component does not necessarily require changes in others. As a result, modules can be upgraded, replaced, or exchanged independently (Veenstra et al., 2006).

In contrast, an integral architecture involves a more complex relationship between functions and components. Several functions may be combined within a single part, or one function may be distributed across multiple parts (Ulrich, 1995). Interfaces are coupled, meaning that changes to one component often affect others and may require coordinated redesign (Veenstra et al., 2006).

Choosing between modular and integral architectures involves trade-offs (Yin et al., 2014). Modular structures support product variation and component standardization, as modules can be recombined to create different product configurations (Muffatto & Roveda, 2002; Ulrich, 1995). However, modular designs may be less effective when optimizing system-level characteristics. Integral architectures, by contrast, can provide better overall performance in attributes such as size, mass, or aesthetics, but they require tighter coordination across components (Ulrich, 1995; Yin et al., 2014).

Standardized and decoupled interfaces are therefore central to modularity (Ulrich, 1995). When interfaces are clearly defined, components become interchangeable, which enables the development of product platforms where shared modules form the basis for multiple product variants (Muffatto & Roveda, 2002). In the housebuilding industry, methods have been developed to measure the degree of coupling between building modules in order to balance standardization and customer variation (Veenstra et al., 2006).

2.1.3 Modular Product Platforms in Housebuilding

A product platform refers to a set of shared components, modules, processes, and knowledge that are used across a family of related products (Muffatto & Roveda, 2002). The purpose of a platform is to balance standardization and variation, enabling firms to achieve production efficiencies while still offering different product configurations (Veenstra et al., 2006). This approach relies on a modular product architecture, where standardized and decoupled interfaces allow modules to be reused and recombined into different variants (Muffatto & Roveda, 2002).

A central concept in platform development is commonality, meaning that components or modules are shared across multiple products (Fixson, 2005). Reusing predefined modules can reduce development time, lower production costs through higher volumes of shared parts, and simplify supply chain management (Dahmus et al., 2001; Fixson, 2005). At the same time, a platform constrains design freedom. Products built on the platform must follow its established interfaces and design rules, which can limit architectural or aesthetic variation (Muffatto & Roveda, 2002).

In the housebuilding industry, platform-based approaches have been introduced to respond to increasing demand for customized housing (Veenstra et al., 2006). Rather than producing identical units, firms use shared structural systems and core utilities as standardized elements, while allowing other components, such as facade treatments or interior layouts, to vary. Methods have been developed to determine which building elements should remain common and which should be variable in order to balance efficiency and customer preferences (Veenstra et al., 2006). This approach supports mass customization but requires careful management of the tension between shared components and design flexibility.

2.1.4 System Integration and Interdependencies

Selecting a product architecture influences not only the product design but also the manufacturing process and the structure of the supply chain (Fixson, 2005). The management of interdependencies between modules, production processes and external suppliers is handled by the system integrator, whose role varies depending on the chosen architecture (Ulrich, 1995).

In a modular architecture, system-level design emphasizes early definition and standardization of component interfaces. This allows development teams or suppliers to work on separate

modules in parallel, since changes within one module do not necessarily affect others. The integrator's role in this context is to ensure that modules comply with predefined interface specifications and function together as intended (Ulrich, 1995).

In contrast, an integral architecture requires continuous coordination across a multidisciplinary team because components share coupled interfaces. Changes to one component can affect several others and may require coordinated redesign. Such interactions influence overall system characteristics, including attributes such as size, weight or aesthetic coherence (Yin et al., 2014). Here the integrator focuses on managing trade-offs and maintaining system functionality as the design develops (Ulrich, 1995).

These structural differences also affect supply chain organization. Modular designs with standardized and decoupled interfaces facilitate outsourcing and simplify inventory management. Integral designs with tightly coupled interfaces often require closer collaboration with suppliers or in-house production to maintain performance and quality (Fixson, 2005; Muffatto & Roveda, 2002). Overall, the chosen architecture shapes the complexity of system integration and the extent to which changes in one component influence the wider system.

2.2 Decision-Making in Complex Engineering Design

2.2.1 Rationality under Constraints

Classical decision theory assumes that decision-makers possess complete information and the capacity to evaluate all available alternatives in order to identify an optimal solution. Under this assumption, individuals are expected to maximize a defined objective by systematically comparing the outcomes of different alternatives (Simon, 1955).

Simon (1955) argued that such assumptions rarely hold in real-world decision contexts. Instead, decision-makers operate under conditions of bounded rationality, meaning that their decisions are constrained by limited information, limited cognitive capacity and restricted time for analysis. Because individuals cannot process all available information or evaluate every possible alternative, the idea of perfectly rational optimization becomes unrealistic in many complex decision situations. As a result, decision-makers typically rely on simplified decision procedures. Rather than identifying the theoretically optimal alternative, they search for solutions that satisfy a set of acceptable criteria. Simon describes this behaviour as satisficing, where alternatives are evaluated sequentially until a solution that meets the decision-maker's requirements is found (Simon, 1955).

The limitations of rational optimization become even more pronounced in complex systems. Simon (1962) demonstrates that many real-world systems are composed of multiple interacting subsystems organized in hierarchical structures. Because interactions between system components generate a large number of possible configurations, exhaustive evaluation of all alternatives becomes increasingly impractical as system complexity grows. Consequently, decision-making in complex environments must rely on simplified decision processes that allow feasible solutions to be identified despite informational and cognitive constraints (Simon, 1962).

2.2.2 Trade-offs in Engineering Design

Engineering design problems rarely involve a single objective. Instead, decision-makers must evaluate alternatives across multiple criteria, such as cost, technical performance and environmental impact. These criteria are often interdependent and may conflict with one another, requiring decision-makers to consider several performance dimensions simultaneously when selecting among alternatives (Triantaphyllou, 2000). Project resources such as time, budget and technical capacity are inherently limited, meaning that improving performance in one area may require compromises in another. As a result, engineering decisions frequently involve unavoidable trade-offs, where improvements in one dimension may reduce performance or limit improvements in others (Tran, 2020).

Sustainability considerations in construction projects further illustrate the presence of trade-offs in engineering decisions. Measures designed to reduce operational energy consumption in buildings often require additional materials or technologies, which may increase embodied environmental impacts. Consequently, actions that improve energy efficiency during the operational phase can lead to higher environmental burdens during material production, emphasizing the importance of assessing impacts across the entire building life cycle (De Oliveira Fernandes et al., 2021).

Trade-offs also arise between environmental performance and economic constraints. Studies evaluating sustainability in construction projects show that solutions with improved environmental outcomes frequently involve higher initial investment costs. This creates a tension between financial feasibility and sustainability objectives in project decision-making (Hussain & Hussain, 2023).

2.2.3 Multi-Criteria Decision-Making

Given the presence of multiple competing criteria in engineering design, structured decision frameworks are required to support the systematic evaluation of alternatives. Multi-Criteria Decision-Making (MCDM) methods provide tools for analyzing problems where several potentially conflicting objectives must be considered simultaneously. These methods structure the decision problem by defining relevant evaluation criteria and enabling alternatives to be assessed within a transparent analytical framework (Velasquez & Hester, 2013).

In construction and sustainability contexts, material selection often requires evaluating criteria such as environmental impact, cost and technical performance. MCDM approaches enable these dimensions to be assessed simultaneously, allowing decision-makers to systematically compare alternative materials and identify trade-offs between competing sustainability objectives (Akadiri et al., 2013; Yaghmour Abdulrahman et al., 2025)

Hybrid MCDM frameworks further extend this approach by combining multiple analytical techniques, enabling the evaluation of both quantitative and qualitative decision criteria in sustainability assessments (Mathiyazhagan et al., 2019). Such approaches allow decision-makers to integrate environmental, economic and technical indicators when comparing alternative materials. In more advanced applications, MCDM methods have also been combined with system dynamics modelling to capture interactions between sustainability indicators over time, providing a more comprehensive basis for evaluating long-term impacts within the built environment (Francis & Thomas, 2023)

2.3 Life Cycle Assessment and Embodied Carbon

2.3.1 System Boundaries and Life Cycle Modules

Life Cycle Assessment (LCA) is a standardized methodology for quantifying environmental impacts across a product system's life cycle (ISO, 2006b). A central methodological choice in any LCA is the definition of the system boundary, which determines which life cycle stages, processes, and flows are included in the analysis and which are excluded (Finnveden et al., 2009). System boundaries are closely linked to the goal and scope of the study and can significantly influence results. In whole-building LCA, boundary definition is frequently identified as a major source of uncertainty, since different inclusions and exclusions may alter calculated environmental impacts (Feng et al., 2022).

Within the building sector, international standards such as ISO 14040 and ISO 14044, together with the European standard EN 15978, provide a modular framework for structuring system boundaries across a building's life cycle (ISO, 2006a, 2006b; Nwodo & Anumba, 2019). This framework divides the life cycle into standardized modules that support consistent reporting and comparison. The product stage, A1–A3, includes raw material extraction, transport to manufacturing facilities, and product manufacturing. The construction stage, A4–A5, covers transport to the construction site and installation processes. The use stage, B1–B7, includes maintenance, repair, replacement, refurbishment, and operational energy and water use, while the end-of-life stage, C1–C4, includes deconstruction or demolition, transport of waste, waste processing, and final disposal. Module D accounts for potential benefits and burdens beyond the system boundary, including avoided impacts from reuse, recycling, or energy recovery in subsequent life cycles (Ly et al., 2025).

Within this modular structure, environmental impacts are commonly distinguished as embodied or operational (Nwodo & Anumba, 2019). Embodied impacts are mainly associated with material production and construction phases, typically corresponding to modules A1–A5, while operational impacts are mainly associated with energy and water consumption during the use phase, particularly module B6 (Ly et al., 2025). Consequently, recent building LCAs increasingly focus on identifying material-related hotspots in early life cycle stages (Feng et al., 2022; Sigurðardóttir et al., 2023).

In material-oriented LCA studies, system boundaries are often limited to a cradle-to-gate perspective, typically covering modules A1–A3 and sometimes extended to A4–A5 (A. S. Jayawardana et al., 2021; Le et al., 2024). This approach excludes use and end-of-life stages and concentrates on impacts associated with material extraction and manufacturing. Cradle-to-gate assessments are frequently applied in comparative analyses of alternative materials, particularly in early design contexts. However, such boundaries do not capture impacts related to maintenance, replacement, demolition, or potential benefits from reuse and recycling, making a clear definition of the selected system boundary essential for correct interpretation of results (ISO, 2006a; Le et al., 2024).

In modular construction systems, system boundary selection requires particular attention. Modular construction relocates significant parts of the construction process from the building site to factory environments, thereby redistributing impacts within modules A1–A5 (Kamali & Hewage, 2016; Ly et al., 2025). Modular systems may also allow disassembly and reuse across multiple use cycles, which increases the relevance of accounting for impacts beyond the initial life cycle. At the same time, many comparative studies focus on pre-use or cradle-to-gate stages in order to isolate embodied impacts associated with material production, transport, and assembly (Finnveden et al., 2009; Sigurðardóttir et al., 2023).

2.3.2 Embodied Carbon in the Built Environment

Embodied carbon refers to the GHG emissions associated with the energy use and processes required to extract raw materials, manufacture building products, transport them, and assemble them into a building. Depending on the scope of the study, embodied carbon can be assessed under different system boundaries, such as cradle-to-gate, covering modules A1–A3, cradle-to-grave, or cradle-to-cradle, which determines which life cycle stages are included (A. S. Jayawardana et al., 2021).

As buildings have become more energy-efficient due to improved envelopes and building systems, the relative importance of embodied carbon has increased, shifting attention from operational performance alone toward full life-cycle environmental impacts (A. S. Jayawardana et al., 2021). In low-energy buildings, embodied emissions may account for a substantial share of total life-cycle emissions (Feng et al., 2022; Salati et al., 2024). Comparable patterns have been observed in modular building research, where the production and construction phases represent a growing share of total impacts as operational energy demand decreases (Kamali & Hewage, 2016).

Empirical building LCAs identify material production and upfront construction, commonly associated with modules A1–A5, as major contributors to embodied emissions (Sigurðardóttir et al., 2023). Reviews of construction-material LCAs highlight that conventional mineral-based materials account for a substantial share of life-cycle GHG emissions in buildings and infrastructure (Barbhuiya & Das, 2023). Previous embodied carbon assessments further indicate that environmental impacts are often concentrated in specific materials or life cycle stages, underscoring the importance of structured prioritization when analyzing complex building systems (Yan et al., 2025; Zhang et al., 2024).

At the same time, LCA studies indicate considerable potential for reducing embodied carbon through alternative material choices and design strategies. Substituting carbon-intensive materials with lower-impact alternatives can substantially reduce the climate impact of building structures and envelopes (A. S. Jayawardana et al., 2021; Sigurðardóttir et al., 2023). Research on circular construction suggests that lower-impact and more circular material alternatives can contribute to reducing embodied emissions, particularly when combined with circular end-of-life strategies (Barbhuiya & Das, 2023; Le et al., 2024).

The robustness of embodied carbon assessments depends on the quality of the input data used in the calculation. Data quality is typically evaluated based on criteria such as technological, geographical and temporal representativeness, as well as completeness and transparency of modelling assumptions (Finnveden et al., 2009). In building LCAs, foreground data usually include project-specific material quantities, while background data are commonly obtained from life-cycle inventory databases or Environmental Product Declarations (EPDs), which provide emission factors for materials and processes (A. S. Jayawardana et al., 2021). Since embodied carbon results are used to compare and prioritize materials, transparent treatment of data quality is essential for ensuring consistent and reliable assessments of material alternatives (Feng et al., 2022; Finnveden et al., 2009).

2.4 Circularity as a Decision Criterion

2.4.1 Circular Economy Principles in Industrialized Construction

In modular and prefabricated construction, circular economy principles are closely linked to standardization, modularization and off-site manufacturing. These characteristics can support more efficient material use, reduce construction waste and create better conditions for disassembly, reuse and component recovery compared with conventional construction processes (Garusinghe et al., 2023; J. Jayawardana et al., 2023; Minunno et al., 2018). Modular buildings can therefore be understood not only as finished products, but also as systems of components that may be maintained, replaced, reused or recovered across multiple life cycles.

Design for Disassembly (DfD) is central to this circular potential. In modular construction, clearly defined layers, reversible connections and accessible interfaces can facilitate the separation of components by material type at the end of life. This enables materials and components to remain in higher-value loops rather than being downcycled, incinerated or sent to landfill (Dams et al., 2021; Minunno et al., 2018). However, the extent to which this potential can be realized depends on how materials are selected, connected and integrated into the modular system.

Several circular construction frameworks emphasize practical criteria such as modularity, simplicity, material transparency, mechanical connections, durability and the possibility of reuse or recycling (Balasbaneh et al., 2025; Keena et al., 2022). These criteria show that circularity is not only determined by the material itself, but also by its function, location and relationship to surrounding components. In modular construction, material choices therefore influence both immediate performance and future recovery, reuse and substitution potential.

2.4.2 Navigating the Carbon-Circularity Trade-off

Although circularity and embodied carbon reduction are often closely related, they do not always lead to the same material choice. A material with low production-stage emissions may still have limited circular potential if it is difficult to disassemble, reuse or recycle at the end of life. Conversely, a material with higher upfront embodied carbon may offer stronger circular performance if it is durable, reusable and supported by established recycling systems (Antwi-Afari, Ng, Chen, et al., 2023; Keena et al., 2022; Yang et al., 2024).

This trade-off is particularly relevant in modular construction, where materials are expected to function within standardized systems and potentially across several use cycles. Bio-based materials may reduce production-stage emissions, but their circular performance can be limited by composite structures, coatings, adhesives or connections that make separation and recovery more difficult (Keena et al., 2022; Z. Li et al., 2024; Santos et al., 2024). In contrast, materials such as steel may have higher initial emissions but can offer greater reuse and recycling potential when designed for disassembly and repeated use (Antwi-Afari, Ng, Chen, et al., 2023; Yang et al., 2024).

As a result, optimizing for embodied carbon alone may overlook important long-term circularity considerations. Material substitution in modular construction should therefore consider both immediate climate impact and the ability of materials or components to remain in high-value loops over time. This makes circularity a distinct but interacting decision criterion alongside embodied carbon, technical feasibility and cost.

2.4.3 Circularity in the Multi-Criteria Selection Logic

Given the trade-offs between carbon reduction and circularity discussed in the previous section, circularity cannot be treated as an isolated objective in material decision-making. Instead, it must be understood as one of several interacting dimensions that influence how materials are evaluated, particularly in complex systems such as modular construction. Existing research shows that material performance varies significantly across environmental, economic and technical criteria, and that no single material consistently outperforms others across all dimensions (Antwi-Afari, Ng, & Chen, 2023; Balasbaneh et al., 2025).

In modular construction, this multidimensionality is further amplified by the system's reliance on prefabricated elements, standardized components and repeated use cycles. Studies demonstrate that materials such as timber, steel and concrete each offer distinct advantages depending on the performance dimension considered. Timber solutions often achieve lower embodied carbon during production, while steel systems tend to offer higher durability, reusability and compatibility with established recycling infrastructures (Antwi-Afari, Ng, & Chen, 2023; Keena et al., 2022; Yang et al., 2024). These differences illustrate that material performance is context-dependent and cannot be reduced to a single optimization criterion.

Within this context, circularity plays a critical role in shaping how materials are assessed beyond initial environmental impact. Design strategies such as DfD, modular standardization and the use of reversible connections influence whether materials can be maintained in high-value loops or become constrained to downcycling or disposal pathways (Dams et al; Dewagoda et al., 2024). As a result, early design decisions determine not only immediate performance but also long-term recovery potential and future substitution opportunities.

Rather than identifying universally optimal materials, these findings highlight the importance of structuring decision-making in a way that acknowledges system complexity and competing objectives. In practice, this means that material evaluation should support informed judgement after high-impact elements have been identified, rather than attempting to optimize all alternatives simultaneously. Decision-support approaches, including multi-criteria frameworks, can assist in comparing alternatives across multiple dimensions, but their role is secondary to the initial prioritization of impactful system components (Kayaçetin et al., 2023; Seddiki & Bennadji, 2025).

2.5 Prioritization Approaches in Complex Systems

2.5.1 Pareto-based prioritization and hotspot logic

In complex systems, resources for analysis, design, and control are always limited. Systems thinking therefore does not primarily aim to "see everything", but to identify the few factors where a small effort produces a disproportionately large system effect. This rationale for prioritization rests on the principle of factor sparsity: in many technical, organizational, and socio-technical systems, only a limited number of variables, components, or subsystems drive most of the outcome, while the remainder contribute only marginally. Similar observations underpin modern algorithms for multi-objective optimization and feature selection, where Pareto-optimal solutions are characterized by sparse solution vectors, only a few decision variables are non-zero and effectively determine the whole solution (Ducci et al., 2025; Lapucci & Mansueto, 2024).

Juran articulated this principle in the classic dichotomy of "the vital few vs. the trivial many", (Juran, 1988) arguing that industrial quality management should focus on the few root causes

that account for the majority of defects or costs, rather than distributing resources evenly across all possible failure modes. The same logic appears in Koch's popularization of the 80/20 principle (Koch, 2011), which describes non-linear relationships between effort and outcome as a general systems pattern: roughly 20% of causes often generate about 80% of results, whether in revenues, defects, or time losses. Koch emphasizes that the Pareto distribution is not an exact law but a heuristic for managing asymmetries in complex systems, and thus an argument for systematic prioritization rather than uniform optimization (Koch, 2011).

In the building and climate context, Pareto-like patterns have been empirically demonstrated. Studies of buildings' climate impact show that a limited set of life-cycle stages, building elements, or material categories account for most greenhouse-gas emissions, strongly supporting a hotspot-based strategy (M. Li et al., 2022; Ostermeyer et al., 2013; Victoria & Perera, 2018). At building-stock and district scale, spatial analyses can identify sub-stocks or geographic zones where emissions, energy use, or other environmental impacts are heavily concentrated, enabling targeted renovation strategies instead of evenly distributed measures (M. Li et al., 2022). In broader environmental and climate-risk analysis, similar hotspot mapping is used to combine multiple stressors (e.g. urban heat island effects, air quality, and vulnerable populations) and direct interventions to those locations in the system where combined risk is highest (Rezvani et al., 2023).
The methodological link between Pareto-based prioritization and hotspot analysis can be understood as an operationalization of systems thinking: first, the system's components and relationships are modelled; second, their relative contributions are quantified; and finally, the nodes or subsets are identified where interventions have the greatest potential system-level effect (Giannelos et al., 2024; Rezvani et al., 2023; Victoria & Perera, 2018). Causal-loop diagrams, network centrality metrics, or cross-impact matrices are then used to find variables that both strongly influence and are strongly influenced by many others, in other words, the system's "vital few" (Giannelos et al., 2024; Victoria & Perera, 2018),

2.5.2 Pareto logic as a theoretical basis for hotspot analysis

Hotspot analysis is an interpretive step within LCA used to identify the processes, materials, life-cycle stages or impact categories that contribute disproportionately to the overall environmental burden of a product system. By isolating these high-impact elements, hotspot analysis enables improvement efforts to be directed where they are most likely to generate meaningful reductions (A. S. Jayawardana et al., 2021), thereby supporting the use of LCA as a decision-support instrument rather than only as a descriptive accounting tool (Russell-Smith & Lepech, 2015). Pareto-based prioritization deepens this method by explaining why such focusing is rational in complex building systems. The conceptual relationship between hotspot analysis and Pareto-based prioritization is summarized in Table 1.

Pareto-informed rationale for hotspot analysis

Aspect of hotspot analysis	**Underlying Pareto / factor-sparsity logic**	**Citations**
Choice of indicators (e.g. climate impact, energy)	A few indicators capture most relevant risks/benefits; the rest add marginal information.	(M. Li et al., 2022; Rezvani et al., 2023; Victoria & Perera, 2018)
Identification of hotspots (elements, modules, zones)	A small subset of components or areas dominates total impact.	(Chen & Li, 2023; M. Li et al., 2022; Ostermeyer et al.,

		2013; Victoria & Perera, 2018)
Prioritisation of measures	Resources are directed to "the vital few" where marginal benefit per invested unit is highest.	(Ducci et al., 2025; Giannelos et al., 2024; Victoria & Perera, 2018)
Iterative model updating	Once major contributors are reduced, new "vital few" emerge among remaining factors.	(Ducci et al., 2025; Lapucci & Mansueto, 2024; Shen, 2024)

Table 1: Conceptual links between Pareto logic and hotspot analysis in buildings

As shown in Table 1, hotspot analysis can be understood as an operationalization of Pareto logic in environmental assessment. The selection of indicators, identification of dominant contributors, prioritization of measures and iterative reassessment all reflect the idea that a limited number of factors often account for a disproportionate share of system-level impact. In this sense, hotspot analysis provides a practical method for translating factor-sparsity logic into material prioritization.

In building and construction LCAs, hotspot analysis is commonly used to identify material groups, components or life-cycle stages that dominate embodied carbon results. Previous studies show that impacts are often concentrated in specific structural elements, envelope systems or material categories, particularly in early life-cycle stages (Sigurðardóttir et al., 2023). Similar patterns have been identified in modular and prefabricated building systems, where climate and resource use may be concentrated in particular module types, connection details or material categories (Kamali & Hewage, 2016; Yan et al., 2025; Ostermeyer et al., 2013; Victoria & Perera, 2018). This reinforces the value of factor-sparsity-oriented design and management, where analytical effort is directed toward the limited set of contributors with the greatest system-level relevance.

However, hotspot results depend on methodological choices such as system boundaries, emission factors and data quality. A cradle-to-gate assessment may highlight different contributors than a cradle-to-grave assessment, and changes in environmental data may affect the relative ranking of materials. Transparent documentation of data sources and system boundaries is therefore important when hotspot analysis is used as a basis for material prioritization (Finnveden et al., 2009; Le et al., 2024; Mahmood et al., 2022).

3 RESEARCH METHOD

3.1 Research Approach and Design

The purpose of this study is to develop a structured approach for reducing embodied carbon in modular building systems through systematic material prioritization and evaluation. To achieve this objective, the study adopts an abductive research approach.

An abductive research approach involves an iterative process where theoretical knowledge and empirical observations are continuously combined during the research process (Säfsten & Gustavsson, 2024). This approach is commonly used when developing or refining analytical frameworks based on both existing theory and empirical insights. In this study, literature related to embodied carbon, material selection and sustainability in construction was used as a basis for developing the proposed approach. Empirical findings from the case study were then used to refine and evaluate the approach.

The study applies a mixed-method research design, combining both quantitative and qualitative methods. Mixed-method approaches allow researchers to integrate numerical analysis with qualitative insights in order to obtain a more comprehensive understanding of complex research problems (Säfsten & Gustavsson, 2024). In this study, quantitative analysis was used to evaluate the environmental impact of materials by combining material quantities provided by the case company with environmental impact data obtained from climate databases and EPDs. Qualitative data was collected through semi-structured interviews with representatives from the case company.

The interviews served two purposes in relation to the study's research questions. First, they provided insights into current material selection practices, technical constraints, implementation conditions and circularity considerations within the organization knowledge that could not be derived from quantitative data alone. Second, the interview findings were used to inform the development of the multi-criteria evaluation framework, by identifying relevant evaluation criteria and informing the weighting structure based on the priorities expressed by respondents across different functional roles.

Empirically, the research is conducted as a single case study focusing on an existing modular building system provided by the case company. Case study research is particularly suitable when investigating complex phenomena within their real-life context and when multiple sources of data are used (Säfsten & Gustavsson, 2024). The case company provided internal data describing the materials used in the building system together with the quantities associated with each material. This data formed the basis for the environmental impact assessment carried out in the study.

3.2 Research Procedure

The research procedure was designed to address the two research questions by developing a structured approach for systematic material substitution in existing modular building systems. Rather than following a strictly linear sequence, the procedure was iterative in nature, reflecting the abductive research approach described in Section 3.1. Theoretical knowledge, quantitative data and qualitative empirical findings were continuously revisited and refined throughout the research process as new insights emerged, as illustrated in Figure 1.

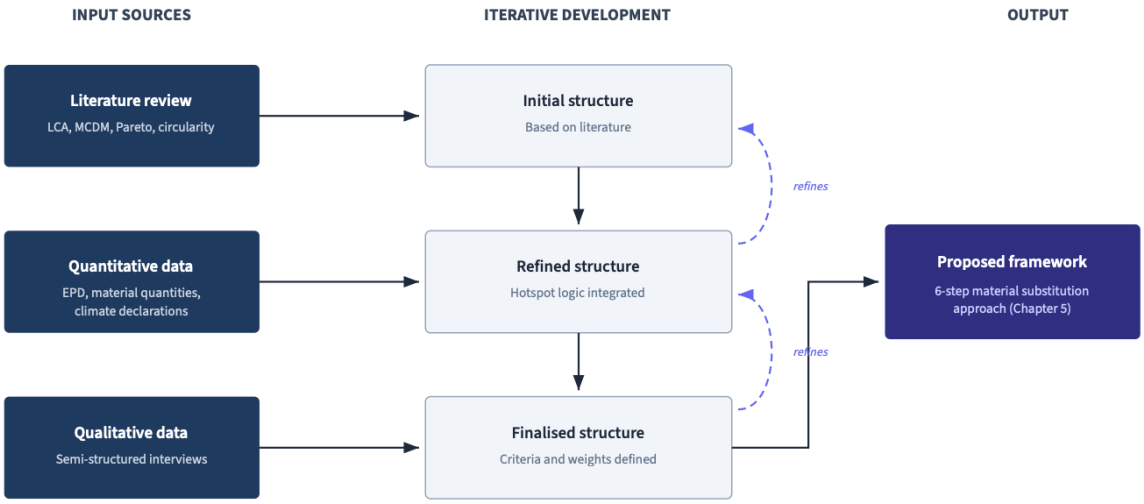


Figure 1. Overview of the iterative research methodology.
Note. Own illustration, created with Canva

The first research question: how high-impact materials can be systematically identified and prioritized, required a procedure capable of translating complex material inventories into structured, comparable environmental data. This was addressed through a combination of document analysis and quantitative calculation, where material quantities from company documentation were combined with emission factors from EPDs and climate databases. A hotspot analysis based on Pareto logic was subsequently applied to identify the materials contributing most significantly to total embodied carbon.

The second research question: how potential material alternatives can be evaluated across embodied carbon, technical feasibility and circular performance, required a procedure that could integrate quantitative environmental data with qualitative organizational knowledge. This motivated the inclusion of semi-structured interviews as a complementary data source, since technical constraints, implementation priorities and circularity considerations could not be derived from EPD data alone. The interview findings were used iteratively to define evaluation criteria and inform the weighting structure of the multi-criteria evaluation framework.

The theoretical framework was developed in parallel with the empirical work and was continuously adjusted as the analysis progressed. Preliminary framework structures were evaluated against both the empirical findings and the theoretical basis and refined until a coherent and structured approach was established. The detailed procedures for data collection and data analysis are described in Sections 3.3 and 3.4 respectively, while the resulting framework is presented in Chapter 5.

3.3 Data Collection

The study relies on multiple complementary sources of data, collected through both document analysis and interviews. This combination reflects the mixed-method design described in Section 3.1 and enabled the research to integrate quantitative data with qualitative organizational knowledge.

3.3.1 Preliminary Study and Document Analysis

Prior to the main data collection, a preliminary study was conducted in collaboration with the company supervisor. This involved both on-site meetings and continuous email correspondence throughout the study period. The purpose of the preliminary study was to establish an understanding of the modular building system, define the scope and limitations of the study, and develop an initial plan for the research. Through this process, access was granted to a range of internal company documents, including the climate declaration of the studied module, technical drawings, product specifications, assembly instructions for project managers, and internal requirement documents. These documents formed the primary quantitative foundation for the embodied carbon assessment and provided essential contextual understanding of the modular product architecture.

3.3.2 Interviews

Semi-structured interviews were chosen as the primary method for collecting qualitative data, as they allow researchers to explore specific topics in depth while still enabling respondents to elaborate on their experiences and perspectives (Alvehus, 2023). A total of six respondents participated across five data collection sessions. Two sessions were individual semi-structured interviews conducted on-site, two sessions were conducted via Microsoft Teams, one individual semi-structured interview and one group interview involving two respondents sharing the same role at different offices. One respondent completed a written questionnaire distributed via email, as they were unable to participate in a physical or online interview. Each interview guide was tailored to the respondent's specific role and area of responsibility. However, all sessions shared the same overarching objective: to develop an understanding of the conditions, constraints and considerations relevant to material substitution within the modular building system. The written questionnaire distributed via email followed the same principle, with questions adapted to the respondent's role and returned as written responses. The interview guides used for each respondent are provided in Appendix 1.

The selection of respondents followed a purposive sampling strategy, where individuals were chosen based on their direct involvement in decisions related to material selection, product development or facility management within the organisation. This approach ensures that the qualitative data captures perspectives from the functions most relevant to the research questions, rather than seeking statistical representativeness across the organisation (Alvehus, 2023). Together, the respondents cover the full decision-making chain relevant to material substitution, from technical requirements and product architecture to sustainability objectives, operational maintenance and procurement conditions. The respondents, their roles and the format of each data collection session are presented in Table 2.

Session	Respondent	Role	Format
1	R1	Product Engineer	Semi-structured interview, Microsoft Teams
2	R2	Renovation Supervisor	Semi-structured interview, on-site
3	R3	Sustainability Manager	Semi-structured interview, on-site
4	R4, R5	Facility Manager (head office),	Group interview, Microsoft Teams

		Facility Manager (southern office)	
	R6	Product Manager	Written questionnaire, email

Table 2: Overview of data collection sessions and respondents.

Although the written questionnaire format did not allow for real-time dialogue, this was not considered a significant limitation in this case. The respondent had served as the primary contact during the preliminary study and had been involved in continuous dialogue with the researchers throughout the data collection period. The written questionnaire was therefore used to complement and consolidate insights already established through prior interaction, rather than as a standalone data source.

All interviews were conducted in Swedish. With the consent of the respondents, interviews were recorded. To support accurate transcription, Microsoft Teams was used to generate real-time transcriptions during the interviews, which were subsequently reviewed and supplemented against the audio recordings to ensure completeness and accuracy. The written questionnaire responses were used directly as text data without transcription.

3.3.3 Ethics

The study was conducted in accordance with established principles for research ethics (Säfsten & Gustavsson, 2024). Prior to each interview, respondents were informed about the purpose of the study, how the data would be used, and that participation was voluntary. Respondents were also informed of their right to withdraw from the study at any time without consequence. Consent was obtained before interviews were recorded.

All respondents are treated anonymously in the report and are referred to by respondent number, R1 to R6, rather than by name. The case company is introduced in general terms in the report, while the specific module series studied is not identified by name. Certain internal documents and data provided by the case company are presented in aggregated form to ensure appropriate handling of organizational information. Interview recordings and transcriptions were stored securely and accessed only by the researchers. Interview data are used solely for the purposes of this study and are not shared with third parties.

Regarding the use of digital tools, Microsoft Teams was used to generate real-time transcriptions during interviews. Respondents were informed of this prior to the interviews. AI-based tools were used in a limited capacity for visualization purposes only, as described in Section 3.5. No AI tools were used to process, analyze or interpret interview data.

The study does not involve vulnerable populations, sensitive personal data or experimental procedures. The research was conducted with respect for the respondents' time, integrity and professional knowledge.

3.4 Data Analysis

The collected data was analyzed using a combination of quantitative and qualitative methods, reflecting the mixed-method design of the study.

3.4.1 Quantitative Analysis: Embodied Carbon Assessment and Hotspot Identification

The quantitative analysis was based on the case company's internal climate declaration spreadsheet, which contained material quantities, initial emission factors sourced primarily from Boverket's database, and pre-calculated total CO_2 values per material. The climate declaration included approximately 55 materials in total. However, 32 of these were identified as reused components and were therefore excluded from the embodied carbon calculations, as reused materials do not generate new production-stage emissions within the A1–A3 system boundary. The quantitative analysis was consequently based on the remaining 23 materials representing new material input to the module.

The embodied carbon of each material was calculated by multiplying material quantity in kilograms by the corresponding emission factor, expressed in kg CO_2e per kg. Total embodied carbon was obtained by summing the individual material contributions. The analysis was limited to life cycle modules A1-A3, corresponding to raw material extraction, transport to manufacturing and product manufacturing, in line with the system boundary defined in Section 3.1.

An initial Pareto analysis was conducted in Microsoft Excel to identify which materials contributed most significantly to total embodied carbon. Rather than applying a fixed percentage threshold, the hotspot identification focused on materials that clearly stood out as dominant contributors within the emission distribution, in line with the factor sparsity logic described in Section 2.5. The purpose of this initial analysis was not to produce final results, but to guide the subsequent data collection by indicating which materials warranted further investigation through product-specific EPDs.

Following the initial screening, product-specific EPDs were collected by contacting the relevant manufacturers directly for the identified high-impact materials. A revised Pareto analysis was then conducted using the manufacturer EPDs, including a cumulative contribution curve, to confirm and refine the hotspot identification. It should be noted that while the same three materials were identified as hotspots in both analyses, the ranking of the second and third materials shifted when manufacturer EPDs replaced the Boverket database values, the material initially ranked third moved to second place, and vice versa. This shift underscores the importance of using product-specific data rather than generic database values, as the choice of emission factor source can influence the relative prioritization of materials even when the overall hotspot set remains the same.

3.4.2 Qualitative Analysis: Thematic Analysis of Interview Data

The qualitative data obtained from interviews and the written questionnaire was analyzed using thematic analysis (Braun & Clarke, 2006). This method supports the systematic identification and interpretation of patterns within qualitative data through an iterative process of coding and categorization.

The analysis began with repeated reading of the interview transcriptions, which had been generated through real-time transcription in Microsoft Teams and subsequently verified against the audio recordings. Recurring patterns related to material selection practices, technical constraints, circularity considerations and implementation priorities were identified and coded. The coded data was then organized into broader themes that captured the central findings across respondents.

The interview findings contributed to the development of the multi-criteria evaluation framework in two ways. First, recurring constraints and priorities expressed by respondents were used directly to identify and define evaluation criteria, for example, technical performance requirements and market availability emerged as explicit priorities across multiple interviews. Second, the relative emphasis placed on different factors by respondents informed the weighting structure of the MCDM model through interpretive analysis. While some criteria emerged directly from respondent statements, the assignment of weights involved an element of researcher interpretation in order to translate qualitative priorities into a structured quantitative framework.

3.5 Research Quality

The quality of the research was evaluated through four criteria commonly applied in mixed-method research: credibility, transferability, dependability and confirmability (Säfsten & Gustavsson, 2024).

Credibility refers to the extent to which the findings accurately represent the phenomenon under study. In this study, credibility was strengthened through triangulation, combining multiple data sources including the company's climate declaration, technical drawings, product documentation, manufacturer EPDs and interview data (Säfsten & Gustavsson, 2024). The use of complementary quantitative and qualitative data enabled cross-checking of information and reduced the risk of drawing conclusions based on a single source. In the qualitative analysis, credibility was further supported by conducting interviews with respondents representing different functional perspectives within the organization, ensuring that the findings reflect a range of organizational viewpoints rather than a single perspective (Braun & Clarke, 2006). Additionally, the revised Pareto analysis using product-specific EPDs rather than generic database values strengthened the credibility of the hotspot identification, as it ensured that the prioritization was based on verified emission data.

Transferability refers to the extent to which the findings and the developed approach can be applied in other contexts (Säfsten & Gustavsson, 2024). As the study is based on a single case study, the empirical findings are primarily applicable to the investigated modular building system and are not intended to be universally generalizable. However, the structured framework developed in the study was designed to be applicable to similar modular construction contexts where systematic material prioritization and substitution evaluation are needed. The transferability of the framework is therefore considered higher than that of the specific empirical findings.

Dependability refers to the consistency and transparency of the research process (Säfsten & Gustavsson, 2024). To strengthen dependability, the calculation procedures and analytical steps were clearly documented throughout the study. The embodied carbon calculations followed a consistent methodology, applying the same system boundary and calculation logic across all materials. The qualitative analysis was conducted systematically using thematic analysis, with transcriptions verified against audio recordings to ensure accurate representation of respondent

statements (Braun & Clarke, 2006). The research procedure is described in sufficient detail in Sections 3.2 through 3.4 to allow the process to be followed and potentially replicated in similar contexts.

Confirmability refers to the extent to which the findings reflect the data rather than researcher bias (Säfsten & Gustavsson, 2024). To strengthen confirmability, the use of standardized environmental data sources such as manufacturer EPDs ensured that the quantitative analysis was grounded in verified third-party data rather than researcher assumptions. In the qualitative analysis, the development of evaluation criteria and weights was based on patterns identified across multiple respondents rather than individual statements, reducing the influence of any single perspective (Braun & Clarke, 2006). Where researcher interpretation was involved, particularly in translating qualitative priorities into criterion weights, this is explicitly acknowledged in Section 3.4.2 and in the framework description in Chapter 5.

Gemini and Canva were used solely as visualization aids to support the creation of figures included in the report. All analytical content, interpretations and conclusions are the researchers' own.

4 EMPIRICAL FINDINGS

4.1 Case Description

4.1.1 Case Company

The case company is a Swedish family-owned manufacturer founded in 1989, specialising in the rental of temporary modular buildings. The company operates across the Nordic region and serves both public and private sector customers, including schools, preschools, offices, care facilities and housing. Modules are manufactured in Sweden using timber as the primary structural material. A defining characteristic of the company's business model is that modules are designed for repeated use, renovated between rental periods and redeployed across multiple customers throughout their service life.

4.1.2 The Studied Module

The module investigated in this study is a modular classroom system designed specifically for primary and secondary school environments. The system features open and flexible floor plan configurations intended to support adaptable learning environments. The module is a standardized product delivered across a large number of projects, meaning that material composition and component specifications remain consistent between deliveries. This standardization makes it particularly suitable for systematic material analysis, as findings are relevant across the entire product family.

4.2 Material Inventory and Embodied Carbon Hotspots

The modular building system comprises multiple material categories. A complete material inventory including material quantities and emission factors is provided in Appendix 2. Figure 2 presents the embodied carbon contributions of all materials included in the module, ranked in descending order. The cumulative curve illustrates the concentration of emissions across the material inventory.

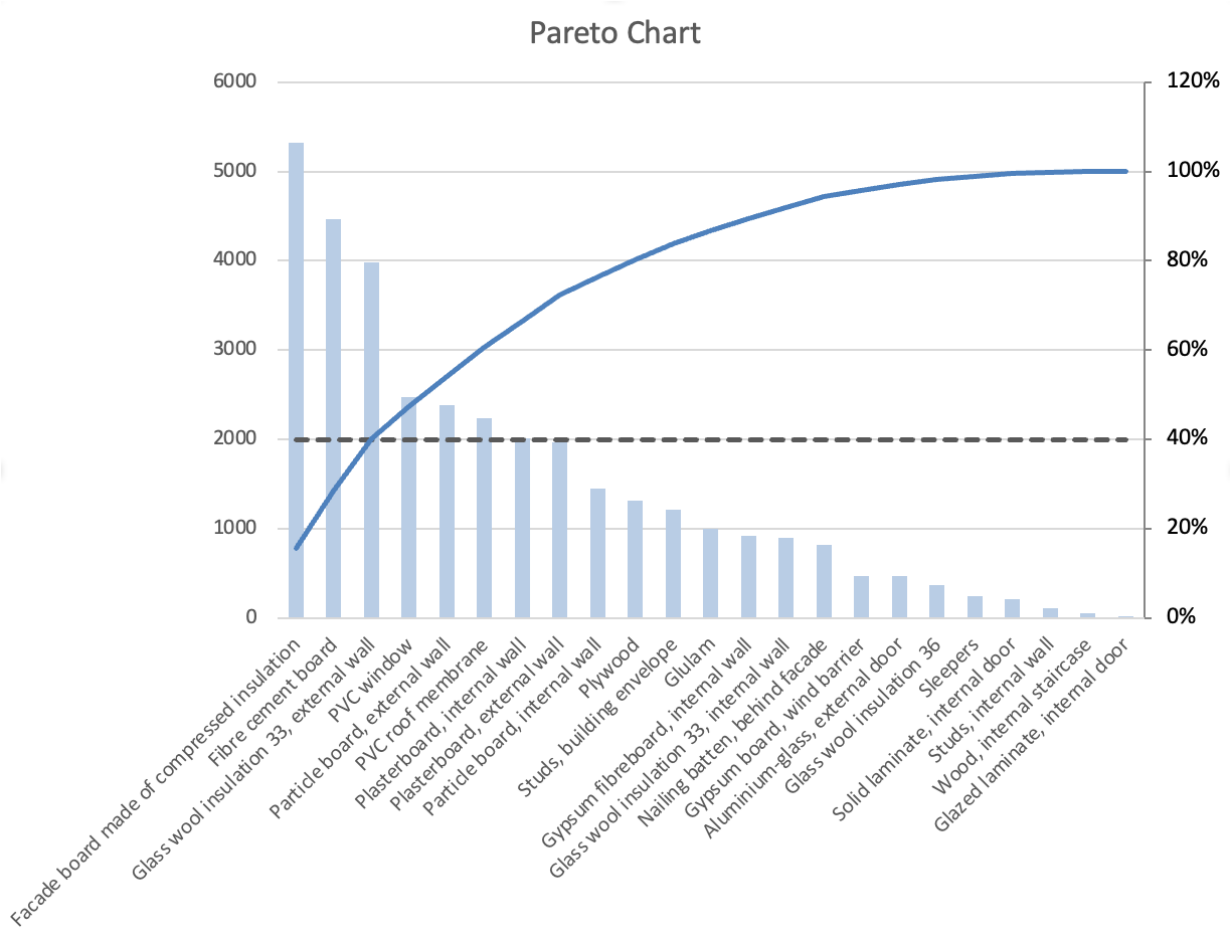


Figure 2. Embodied carbon contributions by material, ranked in descending order, with cumulative share of total emissions.

Three materials were identified as embodied carbon hotspots, collectively accounting for approximately 40% of total emissions: facade board made of compressed insulation, fibre cement board and glass wool insulation 33 in the external wall.

Material	**Quantity (kg)**	**Emission factor (EPD kg CO_2e/kg)**	**Total (kg CO_2e)**	**Share (%)**
Facade board made of compressed insulation	3130,268352	1,7	5321,456198	15,475%
Fibre cement board	6876,3816	0,65	4469,64804	12,998%
Glass wool insulation 33, external wall	4479,529	0,89	2474,304	11,594%
Total top 3				**40,067%**

Table 3: Identified hotspot materials and their respective embodied carbon contributions.

Product-specific Environmental Product Declarations (EPDs) were collected for each of the three identified hotspot materials to obtain verified emission data.

Material	EPD source	System boundary	Emission Factor (EPD kg CO_2e/kg)
Facade board made of compressed insulation	Rockpanel Group	A1-A3	1,7
Fibre cement board	Swisspearl Group	A1-A3	0,65
Glass wool insulation 33, external wall	Saint-Gobain ISOVER	A1-A3	0,89

Table 4: EPD data for the three hotspot materials.

4.3 Interview Findings

This section presents the empirical findings from the conducted interviews. The data has been analyzed using a thematic approach, resulting in three main themes identified across respondents. The findings are structured to describe how material selection is constrained, how decisions are made through trade-offs between multiple criteria, and how decision support is partly structured, combining formal evaluation with experience-based and manual assessment.

4.3.1 System Constraints in Modular Building Systems

Respondent 1, Respondent 2, and Respondent 3 describe that material selection in the modular building system takes place within a set of structural constraints related to technical requirements, compatibility within the modular architecture, and the scale of the existing module fleet. These factors are described as limiting what types of material changes are feasible in practice.

Technical requirements are presented as a fundamental constraint. Respondent 1 and Respondent 2 describe that materials must meet requirements related to fire classification, and that sound and load-bearing performance are also key constraints, though the emphasis varies depending on the material's position in the module and the module series. These requirements limit which alternative materials can be considered, as any option that fails to meet them is excluded.

In addition, the modular structure and its required compatibility impose further constraints. Components must fit within a standardized system where both technical properties and visual appearance remain consistent over time. Respondents describe that new materials need to be compatible with both older and newer modules without affecting the consistency of the system. Respondent 2 emphasizes that modules from different production series must be able to be combined, highlighting the importance of maintaining consistency across the system.

> *"Det är ju otroligt känsliga med att vi ska kunna ställa ihop många modeller med varandra, hur gamla de är och så ska de liksom passa ihop."*
> *"There is a strong sensitivity to ensuring that modules can be combined, regardless of their age, and that they fit together."* (Respondent 2)

The existing module fleet represents an additional constraint on material changes. The company operates approximately 24,000 modules in various configurations, which, according to respondents, makes it difficult to implement material substitutions across the entire system at once. Instead, changes are introduced gradually, often in connection with renovation or maintenance cycles. Respondent 3 highlights that the scale of the existing module base leads to a more incremental approach.

The interviews also describe that certain materials are deeply integrated into the construction, such as insulation embedded within walls, floors, and ceilings. These components require extensive demolition to replace, which involves both time-related and cost-related implications. Similarly, façade materials are described as difficult to handle and replace without significant intervention.

4.3.2 Material Selection as a Multi-Criteria Trade-Off Process

Material selection in the modular building system is characterized as a process of balancing cost, technical function, supplier conditions, and sustainability. Respondents describe how material substitution, particularly for high-emission materials, is shaped by a combination of practical and organizational constraints.

Cost is consistently described as a central limiting factor. Respondent 1 emphasizes that pricing conditions in the market strongly influence material decisions and can act as a barrier to change, while Respondent 3 describes how customer requirements, delivery times, and economic conditions affect how material-related decisions are handled in practice, particularly in relation to the large existing module fleet and the gradual nature of material changes.

> *"...på grund av prisläget i marknaden så är det kostnaden som är väldigt stor... och det kan vara den som sätter käppar i hjulet."*
> *"...due to the pricing conditions in the market, cost becomes very significant... and it can act as a barrier in the process."* (Respondent 1)

Technical performance is, however, treated as non-negotiable. Several respondents describe that materials must meet requirements related to fire safety, structural performance, acoustics, and other building-specific conditions in order to be considered viable alternatives. Materials that fail to meet these requirements are not considered feasible for implementation.

Practical feasibility is also central to decision-making. Respondent 2 emphasizes that materials must be manageable in day-to-day work, meaning that they should not be too heavy and must be possible to handle and install within existing work routines. Respondent 1, in contrast, highlights the importance of standardized products that are readily available on the market, together with supplier-related factors such as delivery capacity, certifications, and reliability.

Respondents from operational and maintenance functions further describe that there is often limited time to compare prices and negotiate with suppliers in detail, making cost, availability, and delivery speed important considerations during maintenance and repair work. Respondent 4 explains that availability is considered more important than handleability in their role, particularly because contractors need quick access to products when issues arise.

> *"Alltså hanterbarheten blir ju inte så viktig i vår roll, så där skulle jag nog säga att det är en tvåa. Tillgängligheten är desto viktigare att våra entreprenörer får ta på produkterna och att de får tag på dem snabbt."*

> *"Handleability is not particularly important in our role, so I would probably rate it as a two on a five-point scale. Availability is much more important, especially that contractors can access products and obtain them quickly."* (Respondent 4)

Respondents also describe how customers often prioritize low cost, while the practical ambition is to provide solutions that are durable and sustainable, creating tensions between price, quality, and sustainability. Technical performance also remains important in use, particularly for installations such as ventilation and other building systems, where functionality is described as directly affecting tenant satisfaction.

Material compatibility with the existing modular system further narrows what is possible. Respondent 2 and Respondent 3 both describe that modules produced at different times or in different locations must still fit together within the same series, meaning that large variations between modules are not acceptable.

Sustainability considerations are also described as being included in material selection decisions, although these are often challenged by other project requirements.

> *"…vi vill ju att det ska alltid beaktas men ibland kommer det lite i konflikt med kundkrav, leveranstider och ekonomi."*
> *"…we want sustainability to always be considered, but it sometimes comes into conflict with customer requirements, delivery times, and economic considerations."* (Respondent 3)

Respondent 6 describes a more structured material selection process in the development of new module series, where materials are evaluated step by step against technical requirements such as fire safety, load-bearing capacity, acoustics, and moisture resistance. The product department is described as proposing materials, after which the factory evaluates buildability and work environment conditions related to handling in production. In cases where materials are difficult to work with, adjustments may be requested before implementation.

The respondent also explains that climate impact is included as one evaluation factor in product assessments together with technical, economic, and practical considerations. At the same time, circularity objectives are described as strongly influencing material selection, where materials associated with landfill disposal are avoided whenever possible. Respondent 6 further highlights that circularity ambitions reduce the number of feasible material alternatives and make the evaluation process more time-consuming, since supplier responses often require manual interpretation to ensure fair comparison and assessment.

4.3.3 Operational Feedback and Experience-Based Decision Support

Respondents describe that material-related decisions are shaped through operational feedback, accumulated experience, and environmental information that is not always easy to compare or apply consistently. Knowledge about materials and material-related problems is described as developing gradually through recurring operational issues, internal discussions, and practical problem-solving.

Respondent 4 and Respondent 5 describe how fault reports and customer complaints during the rental period are registered and documented as deviations in an improvement log. According to Respondent 5, their ability to influence material-related decisions increases when recurring or systematic problems can be identified, whereas isolated incidents caused by handling mistakes

or incomplete documentation are more difficult to address. Material-related improvements are therefore described as emerging gradually through repeated issues in practice, such as components that break down or materials that do not perform well in certain environments.

Respondent 1 similarly explains that material-related decisions are often guided through internal dialogue and accumulated experience, where different factors are considered depending on the situation rather than through a fixed evaluation model. Respondent 3 adds that knowledge regarding material performance and sustainability develops progressively over time as new experiences and operational feedback emerge.

Environmental data is described as being available, but difficult to interpret and apply consistently in practice. Respondent 3 explains that different types of data are used, including both standardized assumptions and supplier-provided information, which makes it challenging to achieve consistency and comparability.

> *"Det är lite blandat schablonberäkningar och leverantörsdata."*
> *"It is a mix of standardized assumptions and supplier-provided data."*
> (Respondent 3)

Respondents also describe uncertainty regarding how environmental data should be defined and applied, particularly in relation to climate declarations and reused modules. Respondent 1 highlights challenges in determining what should be included in such assessments.

> *"…svårt att avgöra vilka grejer som ska vara alltid representerade i en klimatdeklaration… och då kan det vara så att man får ganska missvisande siffror ibland på vad som är en del av det."*
> *"…difficult to determine which elements should always be included in a climate declaration… and this can sometimes result in misleading figures regarding what is actually accounted for."* (Respondent 1)

This uncertainty is described as making it difficult to compare material alternatives and to assess their environmental impact in a consistent way.

At the same time, Respondent 6 describes that certain material-related decisions are supported by a more structured product evaluation process, where climate impact is included as one factor among several evaluation criteria. Materials associated with landfill disposal are also described as being excluded whenever possible due to circularity objectives. The respondent further explains that these evaluations often require manual interpretation and comparison of supplier information, which contributes to making the process partly structured rather than fully standardized.

5 ANALYSIS OF EMPIRICAL FINDINGS

5.1 Prioritization and hotspot identification in modular building systems

The hotspot analysis in this study functions not only as a method for ranking materials, but as an analytical approach for identifying where the greatest reduction potential exists within the modular system. Systems thinking assumes that system outcomes are shaped by the relationships between components, and that analysis should therefore focus on the factors that drive disproportionate system-level effects rather than on all components simultaneously (Arnold & Wade, 2015). In the LCA literature, hotspot analysis is similarly described as a decision-support tool that isolates dominant contributions within complex product systems and thereby enables targeted improvement efforts (A. S. Jayawardana et al., 2021; Russell-Smith & Lepech, 2015; Sigurðardóttir et al., 2023).

The empirical results in this study show such a concentration clearly. The Pareto analysis reveals a distinct break after the three highest-emission materials facade board made of compressed insulation, fibre cement board, and glass wool insulation, which together account for approximately 40 percent of the total embodied carbon of the module. This indicates that emissions are not evenly distributed across the material inventory, but concentrated in a limited number of components that therefore become the most relevant for prioritization. This pattern aligns with previous research showing that embodied carbon is often concentrated in material-intensive elements, particularly in early life cycle stages (Barbhuiya & Das, 2023; Nwodo & Anumba, 2019; Feng et al., 2022).

The hotspot analysis focused on newly introduced materials within the module rather than the full material inventory. While the module consisted of approximately 55 materials in total, 32 reused components were excluded from the embodied carbon assessment, leaving 23 newly introduced materials as the basis for the hotspot analysis. This reflects the study's focus on embodied carbon related to new material production within the selected system boundary and clarifies that the prioritization is based on a defined material subset rather than the complete module composition.

The pattern also reflects Pareto logic and factor sparsity, where a limited number of components account for a disproportionate share of total impact (Arnold & Wade, 2015; Victoria & Perera, 2018). In this sense, hotspot analysis is not only a ranking exercise, but a way of identifying the small set of dominant contributors that warrant further analytical attention within the modular system (A. S. Jayawardana et al., 2021; Russell-Smith & Lepech, 2015; Sigurðardóttir et al., 2023). This also reflects the logic of bounded rationality in complex decision-making, since decision-makers cannot evaluate all possible alternatives simultaneously and therefore rely on simplified processes that reduce complexity to manageable levels (Simon, 1955, 1962).

The hotspot ranking should however be understood as context-dependent rather than as a definitive representation of material importance. The literature emphasizes that results are influenced by system boundaries, data quality, and modelling assumptions, meaning that changes in EPD data or analytical scope may alter the relative ranking of materials (Le et al., 2024; Mahmood et al., 2022; Finnveden et al., 2009). Since this study is limited to A1–A3 stages, the analysis primarily reflects emissions from material production and therefore

highlights materials with high upfront carbon impact within the defined system boundary (ISO, 2006a, 2006b; Ly et al., 2025).

5.2 Material substitution under system constraints

Material substitution in this study cannot be understood as a simple question of replacing one material with another. Product architecture theory shows that modular systems are built through standardized interfaces and functional allocation, where components are connected in ways that shape how changes propagate through the wider system (Ulrich, 1995; Muffatto & Roveda, 2002). This means that material substitution must be assessed as a system-level change, where modifications to one component can influence compatibility, performance, and the functional coherence of adjacent components across the broader product architecture. This is also consistent with the idea of modular product platforms, where common modules and standardized interfaces are used across a product family, meaning that substitution decisions must operate within an established platform logic rather than as isolated one-off changes (Veenstra et al., 2006).

This is especially relevant in the present case because the studied module is part of a standardized product family, where material composition and component specifications are consistent across deliveries. The empirical findings show that material selection is constrained by fire classification, sound performance, and load-bearing requirements, which respondents describe as fundamental and non-negotiable conditions for substitution. Technical feasibility therefore functions as a threshold condition, where materials that do not meet minimum performance requirements cannot realistically be considered for implementation.

The empirical findings also show that some materials are deeply integrated into the construction, particularly insulation embedded within walls, floors, and ceilings. Replacing these components would require substantial disassembly and reconstruction work, which introduces additional time and cost implications and makes material substitution significantly more complex in practice.

This reflects bounded rationality in engineering decision-making, where decisions are made under constraints rather than through full optimization (Simon, 1955). Under bounded rationality, decision-makers cannot evaluate every possible alternative in full, which means that technically unrealistic options are excluded before broader comparisons between alternatives can take place (Simon, 1955; Simon, 1962). In this sense, feasibility reduces the decision space to alternatives that are sufficiently compatible with the existing modular system to warrant further consideration.

The empirical material also shows that feasibility is not only a technical issue, but a question of compatibility across time and across the existing module fleet. Respondents describe that modules must remain compatible within the same product platform, which makes large-scale substitution difficult. The company’s approximately 24,000 existing modules create an additional constraint, since substitutions cannot be introduced across the entire system at once but are instead implemented gradually through renovation cycles, maintenance work, or scheduled upgrades. This means that a technically promising material may still be constrained if it cannot be integrated at the pace and scale required by the existing modular architecture.

The empirical material also indicates that the relevant requirements vary depending on the material’s position within the module and the specific module series, meaning that feasibility is assessed relative to where the component is located within the broader modular architecture.

Supplier conditions further reinforce this interpretation. Respondents emphasize supplier availability, standardized product availability, and delivery reliability as part of feasibility, showing that substitution depends not only on internal technical performance but also on procurement stability and supply chain conditions. Fixson (2005) highlights that product architecture influences not only product design, but also manufacturing processes and the structure of the supply chain. In modular systems, this means that a material may perform well environmentally and technically, but still fail as a realistic option if supply conditions make implementation unstable, delayed, or too costly. In this sense, material substitution affects not only the component itself, but also system integration, production coordination, and supply chain organization.

Technical conditions, supplier availability, and implementation constraints may change over time, meaning that materials that are currently unsuitable may become more feasible as organizational or technical conditions evolve. The findings therefore indicate that material substitution in modular systems is constrained by interdependent technical, organizational, and supply chain conditions rather than by isolated material performance alone.

5.3 Material substitution as a multi-criteria decision problem

Decisions regarding material substitution in modular systems rarely involve a single objective, but instead require simultaneous evaluation of environmental impact, cost, and technical performance, since these criteria are often interdependent and conflicting (Triantaphyllou, 2000; Tran, 2020). Once technically feasible alternatives remain after screening, decision-makers must therefore balance multiple competing objectives rather than optimize a single criterion. From a bounded rationality perspective, it is not possible to fully optimize across all competing criteria simultaneously, which makes trade-offs and satisficing necessary in complex material evaluation processes (Simon, 1955; Simon, 1962).

A further analytical issue concerns comparability between alternatives. Triantaphyllou (2000), Velasquez and Hester (2013), and Akadiri et al. (2013) argue that environmental performance, cost implications, technical requirements, and practical constraints cannot easily be reduced to a single comparable measure, since these criteria are often qualitatively different and partially conflicting. This means that material substitution decisions require simultaneous consideration of multiple dimensions rather than isolated judgement based on a single dominant factor.

Circularity adds an additional layer of complexity to this comparison process, since reuse potential, resource efficiency, and end-of-life performance must be considered alongside environmental and economic performance rather than treated as isolated sustainability goals (Garusinghe et al., 2023). In modular and prefabricated construction, standardization, modularization, and off-site manufacturing enable reuse and disassembly, but circularity and carbon reduction do not always move in parallel (Minunno et al., 2018; J. Jayawardana et al., 2023). As a result, alternatives that reduce embodied carbon are not automatically the same alternatives that maximize reuse potential or long-term circular performance.

The empirical findings show that this trade-off logic is highly relevant in the studied modular building system. Respondents describe material selection as a continuous balance between cost, technical function, supplier conditions, and sustainability, with cost repeatedly identified as the strongest limiting factor. This indicates that embodied carbon reduction is rarely the primary driver of substitution decisions, particularly when alternative materials create higher purchasing costs or increase uncertainty in implementation. Within this context, technically

feasible alternatives must therefore be evaluated in relation to multiple competing objectives rather than environmental performance alone.

At present, material-related decisions are shaped by a mix of standardized assumptions and supplier-provided information, but the interviews show that this data is not always easy to interpret or apply consistently in practice. Some decisions are supported by more structured product evaluations where climate impact is included as one factor among several, yet manual interpretation and comparison of supplier information are still required. Bounded rationality becomes visible in how complex evaluation situations and imperfectly comparable information create a need for simplified procedures and practical judgement in order to make material assessments manageable (Simon, 1955; Simon, 1962). From a multi-criteria perspective, this also shows why competing criteria should be assessed through transparent evaluation structures rather than through isolated judgement or reliance on a single dominant factor (Triantaphyllou, 2000; Velasquez and Hester, 2013; Akadiri et al., 2013).

The scale of the existing module fleet further reinforces this trade-off logic, since implementation pace and compatibility requirements influence how quickly technically feasible substitutions can realistically be introduced. Respondents emphasized that even technically and environmentally attractive alternatives may need to be implemented gradually through renovation cycles, maintenance work, or scheduled upgrades rather than through immediate system-wide replacement. Supplier availability, standardized product availability, and delivery reliability also influence the comparison between alternatives, showing that material decisions are shaped not only by environmental and economic performance but also by procurement and supply chain conditions (Fixson, 2005). In this sense, implementation pace and supplier stability influence which technically feasible alternatives can realistically be introduced within the constraints of the modular system.

Within the studied modular system, material substitution decisions are therefore shaped by trade-offs between environmental ambition, technical feasibility, economic considerations, and implementation practicality.

5.4 From informal judgment to structured decision support

Environmental information and supplier data are not automatically translated into consistent material decisions within the studied modular system. The empirical findings show that this information is not always easy to interpret or apply consistently in practice. Respondents describe how climate considerations are still only partly integrated into structured product evaluations, and that manual comparison of supplier information remains necessary when assessing material alternatives. The interviews further indicate that some decisions are already partly structured through formal product evaluations in which climate impact is included as one factor among several, but that interpretation and comparison of supplier information still depend heavily on manual assessment.

Respondents also describe how operational conditions such as delivery time, supplier reliability, certifications, delivery capacity, and price comparisons influence which alternatives can realistically be considered in practice, particularly in maintenance and renovation situations where implementation speed is important. The large existing module fleet further reinforces these constraints, since substitutions must often be introduced gradually through renovation cycles and ongoing maintenance rather than through immediate system-wide replacement. This means that decisions are frequently made under operational time pressure, where accumulated experience and practical judgement remain useful because they allow faster adaptation to

situational constraints. The data is therefore not lacking, but it is difficult to translate into a consistent decision logic, which means that environmental information does not automatically lead to better or more comparable decisions. As a result, the current decision process depends heavily on operational feedback, accumulated experience, and practical problem-solving rather than on a fully standardized evaluation process.

Bounded rationality is evident in organizational decision-making, where simplified decision procedures and practical judgement become necessary in order to manage complex and uncertain evaluation situations (Simon, 1955; Simon, 1962). While tacit knowledge and operational experience remain important, the interviews also show that this type of judgement becomes difficult to scale across a large modular product architecture, particularly when material decisions must remain compatible with standardized interfaces, existing module conditions, and gradual implementation over time (Ulrich, 1995; Muffatto & Roveda, 2002; Veenstra et al., 2006; Fixson, 2005). Informal judgement may therefore function in isolated cases, but becomes less reliable when the same decision logic must be applied consistently across a large and standardized module fleet.

A further challenge concerns comparability and uncertainty in environmental assessments. The findings indicate that supplier information and embodied-carbon data often require manual interpretation before alternatives can be meaningfully compared, while differences in data quality, modelling assumptions, and supplier information may lead to inconsistent or misleading comparisons between alternatives if uncertainty is not handled transparently (Finnveden et al., 2009; Barahmand & Eikeland, 2022).

The empirical findings also show that circularity ambitions influence which alternatives are considered feasible from the outset. Respondents describe that materials associated with landfill disposal are avoided, when possible, which reduces the number of acceptable alternatives before the comparison process even begins. This narrows the decision space and makes the evaluation process more time-consuming, particularly when supplier responses and environmental information must be manually interpreted and compared. Circularity therefore functions not only as an additional evaluation criterion, but also as a filter that shapes which alternatives are considered practically acceptable within the modular system.

Hybrid MCDM approaches are particularly relevant in this context because they allow environmental, economic, technical, and practical criteria to be evaluated within the same decision structure while also making the basis for comparison more transparent and reproducible across projects (Mathiyazhagan et al., 2019; Francis & Thomas, 2023). Explicitly structured and weighted criteria may therefore reduce reliance on purely local interpretation and tacit knowledge by translating complex evaluation situations into a more consistent and repeatable organizational decision process.

Increasing decision complexity and reliance on manually interpreted information therefore create a growing need for more transparent, reproducible, and scalable approaches to material evaluation within modular systems.

6 PROPOSED FRAMEWORK FOR MATERIAL SUBSTITUTION

6.1 Framework Overview

Based on the embodied carbon assessment, hotspot analysis, empirical findings and analysis, a structured framework was developed to support systematic material substitution in existing modular building systems. The framework is intended to support embodied carbon reduction through structured prioritization, feasibility assessment, multi-criteria evaluation and implementation planning within existing modular product architectures.

The overall input to the framework consists of climate declarations, product-specific EPDs, technical drawings, material inventories, product documentation, technical requirements, supplier information and organizational knowledge from the company. Through six sequential steps, the framework processes this information into prioritized material substitution pathways. These pathways indicate which alternatives can be implemented directly, which require longer-term redesign, which should be monitored and which are currently non-feasible.

As illustrated in Figure 3, the framework consists of six sequential steps: material mapping and system understanding, embodied carbon assessment within the A1–A3 system boundary, hotspot identification and prioritization, substitution feasibility screening, multi-criteria evaluation, and implementation prioritization and iterative reassessment. The framework also includes an iterative feedback loop between implementation outcomes and renewed feasibility screening.

Each step is described in detail in the following subsections, including the required input, the process to be carried out and the expected output.

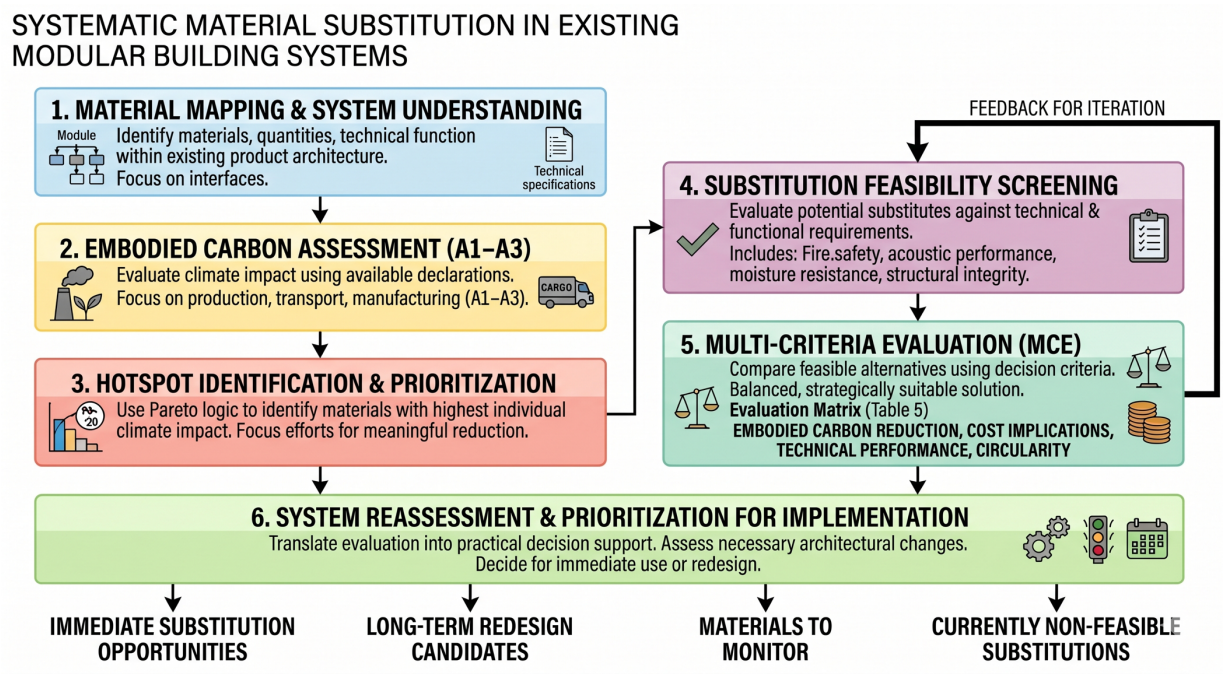


Figure 3. Proposed Framework

Note. Own illustration, created with Gemini

6.1.1 Material Mapping, Carbon Assessment, and Hotspot Identification

The first three steps of the framework focus on identifying the material composition of the modular system and establishing where the largest embodied carbon impacts occur. These steps draw on climate declarations, EPDs and emission factor data, technical drawings, product documentation and material inventories for the studied module.

Material mapping identifies each material together with its technical function, quantity, location and relationships to surrounding components within the modular system. This mapping provides the basis for the subsequent embodied carbon assessment, where emissions associated with each material are quantified within life-cycle modules A1–A3. Emission factors are obtained from product-specific EPDs where available, with generic emission factor data serving as a fallback when product-specific data is not accessible. Material quantities and emission factors are converted into comparable CO_2e values to establish a consistent carbon inventory for the module.

Based on this inventory, hotspot identification is performed using Pareto-based prioritization. Materials are ranked by their total embodied carbon contribution to identify a limited set of materials that together account for a dominant share of emissions. The output of these steps is a prioritized set of hotspot materials that proceed to substitution feasibility screening.

When applied to the case study module, these steps identified facade board made of compressed insulation, fibre cement board, and glass wool insulation as the three dominant hotspot materials, together accounting for approximately 40 percent of the total embodied carbon.

6.1.2 Substitution Feasibility Screening

The fourth step evaluates whether potential alternatives to the identified hotspot materials can realistically be implemented within the existing modular system. This step draws on the prioritized hotspot materials together with technical specifications, certification requirements, supplier information and implementation-related conditions connected to the modular platform.

The screening process assesses potential substitution alternatives against the minimum technical and organizational requirements necessary for implementation. This includes evaluation of fire classification, acoustic performance, structural integrity, moisture resistance, certification compliance, compatibility with surrounding components and interfaces, supplier availability, standardized product availability, delivery reliability and implementation stability. The technical requirements and certification criteria applicable to the three hotspot materials identified in this study are documented in Appendix 3, which provides a structured reference for evaluating whether potential alternatives meet the minimum conditions for substitution.

The screening also considers the conditions of the existing module fleet. Since modular systems may consist of a large number of existing modules, substitutions are expected to be implemented gradually through renovation cycles, maintenance work or scheduled upgrades rather than through immediate system-wide replacement.

Alternatives that satisfy the required conditions proceed to multi-criteria evaluation, while alternatives that do not meet the requirements are excluded from further assessment under current conditions. The output of this step is a set of technically and organizationally feasible substitution alternatives.

6.1.3 Multi-Criteria Evaluation Matrix

The fifth step evaluates the feasible substitution alternatives that passed the feasibility screening. This step draws on the technically and organizationally feasible alternatives together with environmental, technical, economic and organizational evaluation data.

Since all alternatives that reach this stage have already satisfied the minimum technical requirements in the previous step, the technical performance criterion is used to distinguish between alternatives that meet the minimum requirements and those that exceed expectations. The evaluation is performed using the multi-criteria evaluation matrix presented in Table 5, which assesses alternatives across seven criteria: embodied carbon reduction, technical performance, service life, relative cost, circularity and reuse potential, handleability, and market availability.

Criterion	**Definition**	**Score 1**	**Score 3**	**Score 5**	**Source**
C1: Embodied Carbon	EPD emission factor A1–A3 (kg CO_2e/kg) compared to current material	Same or higher CO_2	0–20% lower	>40% lower	EPD data
C2: Technical Performance ⚠️	Meets fire, acoustic and structural requirements of the modular system		Meets the minimum (3)	Fully meets and exceeds (5)	Interview R1 + company technical documentation
C3: Service Life	Estimated functional life within the modular system (years)	<10 years	15–20 years	>30 years	Company evaluation template
C4: Relative Cost	Cost difference compared to current material (%)	>50% more expensive	10–25% more expensive	Cheaper than current	Interview R1/R3 + company evaluation template

C5: Circularity	Disassembly and reuse potential at end-of-life	Landfill or incineration	Recyclable, limited reuse	High reuse potential	Interview R2 + company design documentation
C6: Handleability	Possibility to install within existing work routines and equipment	Requires new equipment or training	Minor adaptation needed	Standard tools, no training required	Interview R2
C7: Market Availability	Standardised product available from multiple reliable suppliers	Single-source specialist produc	Limited number of suppliers	Multiple reliable suppliers	Interview R1 + company evaluation template

Table 5. Multi-Criteria Evaluation Matrix for Material Substitution

Embodied carbon reduction, service life and relative cost are evaluated using predefined quantitative intervals, while technical performance, circularity and reuse potential, handleability and market availability are evaluated using structured qualitative scoring levels. For technical performance, alternatives that satisfy the minimum requirements receive a score of 3, while alternatives that surpass the minimum requirements receive a score of 5.

Criterion weights were assigned based on the empirical findings and are presented in Table 6.

Criterion	**Raw Weight (1–5)**	**Normalised Weight (%)**	**Rationale**
C1: Embodied Carbon	4	16.7%	Primary objective of the study
C2: Technical Performance	5	20.8%	Non-negotiable system requirement: Interview R1
C3: Service Life	3	12.5%	Durability requirement: company evaluation template
C4: Relative Cost	3	12.5%	Key economic constraint: Interview R1 and R3
C5: Circularity / Reuse	3	12.5%	Core sustainability dimension: Interview R2
C6: Handleability	2	8.3%	Practical implementation constraint: Interview R2

C7: Market Availability	4	16.7%	Procurement constraint: Interview R1 + company evaluation template
Total	**24**	**100%**	

Table 6. Criterion weights applied in the multi-criteria evaluation.

The total weighted score for each alternative is calculated according to the equation below.

$$S = \sum_{n=1}^{n} w_i * s_i$$

Where S represents the total weighted score for the evaluated alternative, wi represents the normalized weight assigned to criterion i, si represents the score assigned to the alternative for criterion i, and n is the total number of evaluation criteria.

The framework also includes an interactive spreadsheet-based evaluation tool presented in Appendix 4. The tool includes adjustable criterion weights, evaluation sheets, instructions, and automatically calculated results to support practical application of the framework.

The output of this step consists of evaluated and comparatively assessed substitution alternatives.

6.1.4 Implementation Prioritization and Iterative Reassessment

The sixth and final step translates the evaluated alternatives into practical decision support by categorizing them according to implementation readiness and substitution potential within the existing modular system. This step draws on the evaluated alternatives together with organizational knowledge about implementation conditions, supplier conditions and module fleet constraints.

Based on the evaluation outcomes, alternatives are categorized as immediate substitution opportunities, long-term redesign candidates, alternatives to monitor, or currently non-feasible substitutions. Immediate substitution opportunities represent alternatives that satisfy technical requirements and can realistically be implemented under current organizational and supplier conditions. Long-term redesign candidates represent alternatives that may require interface adjustments, supplier development or broader system modifications before implementation becomes feasible. Alternatives to monitor remain relevant for future reassessment if environmental performance, technical solutions or market conditions change over time. Currently non-feasible substitutions are documented as unsuitable under present constraints but may be reconsidered if technical, supplier or organizational conditions change.

The framework also includes an iterative feedback loop between implementation outcomes and renewed feasibility assessment, illustrated in Figure 3. This allows alternatives to be reassessed if technical conditions, supplier availability, certification requirements or environmental data change over time. The output of this step consists of prioritized implementation pathways for substitution alternatives within the existing modular system.

6.2 Discussion of the Proposed Framework

The proposed framework addresses the identified need for a structured and operational approach to material substitution in existing modular building systems. By combining Pareto-based hotspot identification with multi-criteria evaluation, the framework provides a two-step decision support process that reduces the complexity of material substitution decisions without requiring full building-level life cycle assessment.

6.2.1 The rationale for a two-step structure

A central design choice in the framework is the sequential combination of hotspot identification and multi-criteria evaluation. This structure reflects the bounded rationality constraints described by Simon (1955, 1962), which establish that decision-makers in complex systems cannot evaluate all alternatives simultaneously. Rather than attempting to assess all materials and all possible substitutes at once, the framework first reduces the decision space to the most impactful materials through Pareto logic, and then applies structured evaluation only to those materials that warrant further investigation. This sequencing is consistent with the factor sparsity principle described in Section 2.5, where directing effort towards the vital few produces disproportionately large outcomes relative to the resources invested.

The empirical findings from the present study illustrate this principle clearly. Three materials out of a total of 22 new materials in the module accounted for approximately 40 per cent of total embodied carbon. Attempting to evaluate substitution potential across all 22 materials simultaneously would have distributed analytical effort across contributors of vastly different significance. The Pareto-based first step therefore not only simplifies the decision process but also ensures that substitution efforts are directed where the potential for meaningful carbon reduction is greatest.

6.2.2 Framework contribution relative to existing literature

Existing research on embodied carbon in modular construction has predominantly focused on quantifying environmental impacts at building or system level, with less attention given to how such results can be operationalized into structured material decision processes within ongoing product development contexts, as identified in the problem statement. The proposed framework addresses this gap by translating hotspot analysis results into a practical evaluation structure that integrates environmental, technical, economic and circularity criteria within a single decision support process.

The framework explicitly incorporates organizational constraints, such as the conditions of an existing module fleet, supplier availability and implementation feasibility, as evaluation criteria alongside environmental performance. This reflects the finding from the interviews that material substitution decisions in practice are shaped by a broader set of constraints than embodied carbon alone, consistent with the trade-off perspective described in Section 2.2. By structuring these constraints as explicit and weighted criteria, the framework makes the basis for substitution decisions transparent and reproducible rather than relying on implicit organizational judgement.

6.2.3 Critical reflection on the MCDM component

While the multi-criteria evaluation structure provides a systematic basis for comparing substitution alternatives, several aspects of the MCDM component warrant critical reflection.

The criterion weights were assigned based on the qualitative findings from the interviews, interpreted by the researchers. Although the weighting rationale is documented and grounded in empirical data, the translation of qualitative priorities into numerical weights inevitably involves an element of researcher judgement. Different interpretations of the same interview data could result in different weight distributions, which in turn could influence the ranking of alternatives. This limitation is partially addressed by the interactive spreadsheet tool provided in Appendix 4, which allows users to adjust weights to reflect different organizational priorities and explore how changes in weighting affect the ranking of alternatives. This sensitivity function is considered an important feature of the framework, as it makes the influence of weighting assumptions explicit and supports more informed decision-making.

The treatment of technical performance as a two-stage criterion reflects the empirical finding that technical requirements are non-negotiable within the studied system. Alternatives that fail to meet minimum requirements are excluded during the feasibility screening, while those that pass are further distinguished in the multi-criteria evaluation, where alternatives that exceed minimum requirements receive a higher score. While this design choice is well-supported by the interview data, it means that alternatives failing to meet minimum technical requirements are excluded regardless of their performance on other criteria, including embodied carbon. In some contexts, this could result in the exclusion of materials with significantly lower climate impact that might otherwise be considered through system redesign or interface modification. The framework therefore implicitly prioritizes implementation feasibility within the existing system over maximum carbon reduction potential, which is a deliberate but context-specific design choice.

A further consideration concerns the transferability of the MCDM component. The criterion weights applied in this study are informed by the empirical findings from a specific modular building system and therefore reflect the priorities and constraints of that particular context. Linking the evaluation too rigidly to context-specific technical requirements would risk reducing the framework's applicability in other settings. The two-step structure addresses this by separating the feasibility screening, which is inherently system-specific, from the weighted multi-criteria evaluation. Since the criterion weights are adjustable in the interactive tool, the MCDM component can be reconfigured to reflect different organisational priorities, material contexts or system requirements. This design choice is deliberate, it preserves transferability while still allowing the framework to be grounded in empirically informed priorities when applied to the studied system.

6.2.4 Limitations of the framework

Several limitations of the proposed framework should be acknowledged. First, the framework was developed and applied within a single modular building system and has not been tested against actual substitution alternatives within the scope of this study. The evaluation criteria and scoring logic are theoretically grounded and empirically informed, but their practical applicability and predictive validity remain to be verified through application to real substitution cases.

Second, the system boundary of the embodied carbon assessment is limited to modules A1–A3, which captures production-stage emissions but excludes impacts related to maintenance, replacement, end-of-life treatment and potential reuse benefits. While circularity is included as a qualitative evaluation criterion, the framework does not quantify circular benefits through full life cycle modelling. This means that materials with strong end-of-life reuse potential may be undervalued relative to their full life cycle performance.

Third, reused components were excluded from the embodied carbon assessment, as their production-stage emissions fall outside the defined system boundary. This exclusion means that the framework does not capture the environmental benefit of existing reuse practices within the modular system, which may be significant given the company's business model of repeated module deployment.

Finally, although the framework was designed to be transferable to similar modular construction contexts, its criteria definitions, weight assignments and feasibility conditions reflect the specific organizational and technical context of the studied company. Application in other contexts would require adaptation of the evaluation criteria and weights to reflect the priorities and constraints of the relevant system.

6.2.5 Theoretical alignment

The framework aligns with the theoretical perspectives presented in Chapters 2.1 through 2.5 in several ways. The Pareto-based prioritization reflects the factor sparsity logic described in Section 2.5, where a small subset of materials dominates total embodied carbon. The multi-criteria structure reflects the trade-off perspective in Section 2.2, where engineering decisions involve balancing competing objectives across environmental, technical and economic dimensions. The inclusion of circularity as an evaluation criterion is consistent with Section 2.4, which establishes that circularity and carbon reduction must be considered jointly rather than in isolation. Finally, the system-level perspective embedded in the feasibility screening, which considers interface compatibility, module fleet conditions and supplier availability, reflects the systems thinking principles described in Section 2.1, where material decisions cannot be treated in isolation from the broader product architecture.

7 CONCLUSION AND DISCUSSION

7.1 Conclusion

The purpose of this study was to develop and apply a structured approach for identifying and prioritizing high-impact materials within an existing modular building system, and to evaluate potential material alternatives that may reduce embodied carbon while considering technical feasibility and circularity.

The study addressed the following research questions:

How can high-impact materials be systematically identified and prioritized within an existing modular building system to support reductions in embodied carbon?

How can material alternatives be systematically evaluated across embodied carbon, technical feasibility and circular performance to support substitution decisions within a modular building system?

Regarding the first research question, high-impact materials can be identified and prioritized through embodied carbon assessment followed by Pareto-based hotspot analysis. In the studied module, embodied carbon was concentrated in a limited number of materials rather than evenly distributed across the material inventory. Facade board made of compressed insulation, fibre cement board, and glass wool insulation were identified as the three largest contributors, together accounting for approximately 40 percent of the total embodied carbon of the module. This shows that hotspot analysis can be used as a prioritization tool to identify which materials should be targeted first in material substitution efforts.

Regarding the second research question, material alternatives can be evaluated through a two-step structure consisting of feasibility screening followed by multi-criteria evaluation. Feasibility screening is used to exclude alternatives that do not meet technical or organizational requirements, while multi-criteria evaluation enables feasible alternatives to be compared across embodied carbon, technical performance, cost, supplier conditions, implementation feasibility and circularity. This structure is necessary because material substitution decisions in modular building systems involve several criteria that must be considered simultaneously.

The study addresses the problem identified in the introduction by showing how embodied carbon results can be translated into a structured material decision process. Previous research has mainly focused on quantifying embodied carbon at building or system level, while providing limited operational guidance for how such results can support systematic material decisions in ongoing modular product development. This study shows that hotspot analysis, feasibility screening, multi-criteria evaluation and implementation prioritization can be combined into a structured framework for material substitution in existing modular building systems.

Overall, the findings demonstrate that embodied carbon reduction in modular building systems can be supported through systematic prioritization and structured evaluation. The proposed framework reduces analytical complexity, identifies high-impact materials, and supports

comparison of substitution alternatives while accounting for technical, economic, organizational and circular requirements.

7.2 Discussion

Material substitution in existing modular building systems involves navigating a complex set of environmental, technical, economic and organizational constraints simultaneously. The findings of this study illuminate how these constraints interact in practice and what implications this has for structured decision support in industrialized construction contexts.

One observation that merits broader reflection concerns the role of data quality in material prioritization. In this study, the hotspot analysis was initially conducted using the company's existing climate declaration, in which a large proportion of the materials were associated with generic emission factors from Boverket’s database rather than manufacturer-specific EPDs. Manufacturer-specific EPDs were subsequently collected for the three identified hotspot materials, and the revised analysis showed that the ranking of two materials shifted as a result. While the same three materials remained dominant, this raises a broader question: had the company's climate declaration been based on manufacturer-specific EPDs from the outset, the prioritization might have produced a different result, potentially identifying different materials as the primary targets for substitution efforts. This suggests that the quality of the underlying environmental data is not a neutral input but actively shapes which materials are considered most important to address. For prioritization approaches to be reliable, the data on which they are based must be sufficiently representative of actual product performance rather than generic approximations.

The empirical findings also show that material substitution decisions are shaped by a broader set of constraints than environmental performance alone. Cost and market availability were consistently identified as central limiting factors, where materials from less established or smaller manufacturers may be ranked lower due to limited supply reliability, restricted availability or procurement uncertainty. Sustainability considerations were described as frequently coming into conflict with customer requirements, delivery times and economic conditions. This tension is not unique to the studied context but reflects a wider challenge in sustainable construction, where environmental objectives must compete with operational and economic priorities within existing organizational structures. The relationship between circularity and carbon reduction adds further complexity, as materials with low embodied carbon do not automatically offer high reuse potential and vice versa. Navigating these trade-offs requires decision structures that make competing objectives explicit and comparable rather than leaving them to informal judgement.

The interviews revealed that material-related decisions currently rely heavily on accumulated experience and manual interpretation of supplier information. While this approach is functional in isolated cases, it becomes difficult to apply consistently across a large and standardized module fleet where decisions must remain compatible with established interfaces and platform logic. The proposed framework responds to this by providing a structured evaluation process that incorporates organizational priorities through adjustable criterion weights. However, the framework has not been tested against actual substitution alternatives within the scope of this study, and its practical usefulness therefore remains to be verified through application.

7.2.1 Suggestions for future research

The present study developed and proposed a structured framework for material substitution in existing modular building systems. Several directions are identified for future research to build on and extend the findings of this study.

The most immediate direction concerns the application of the framework to actual substitution alternatives for the three identified hotspot materials: Facade board made of compressed insulation, Cembrit fibre cement board and glass wool insulation. The present study focused on developing and structuring the framework, while the evaluation of specific material alternatives remains to be completed. Future studies could apply the framework to identified substitution candidates in order to test the scoring logic, criterion definitions and weighting structure against real material data and practical outcomes.

A second direction concerns the transferability of the framework to other modular building systems and contexts. The framework was developed within a specific organizational and technical context, and its criteria definitions and feasibility conditions reflect this. Future studies could apply the framework to other module types, building categories or modular manufacturers in order to assess whether the structure and criteria remain applicable across different systems and to identify adaptations that may be needed.

A third direction involves extending the system boundary of the embodied carbon assessment beyond modules A1–A3. The present study focused on production-stage emissions, which provided a consistent and comparable basis for material prioritization. A full cradle-to-grave assessment including use-phase maintenance, replacement cycles, end-of-life treatment and potential reuse benefits would provide a more complete picture of the climate performance of substitution alternatives. This is particularly relevant given the company's business model of repeated module deployment, where end-of-life reuse may significantly influence the total life cycle impact of material choices.

7.2.2 Recommendations to the Company

Based on the findings of this study, the following recommendations are directed to the case company.

It is recommended that the company updates its existing climate declarations with product-specific EPDs from manufacturers rather than relying on generic database values such as those from Boverket. As demonstrated in this study, the choice of emission factor source can influence the relative ranking of materials and therefore the prioritization of substitution efforts. In the present study, the ranking of two hotspot materials shifted when manufacturer EPDs replaced the Boverket database values, illustrating that generic emission factors may not accurately reflect the climate performance of specific products. Systematic collection of manufacturer EPDs for the materials included in the climate declaration would therefore improve the accuracy and reliability of future embodied carbon assessments and provide a stronger and more transparent foundation for material decisions within the product development process.

It is recommended that the company conducts a pilot application of the proposed framework on a specific module type. The framework provides a structured starting point for identifying high-

impact materials and evaluating substitution potential, but its full value is realized through practical application to real material alternatives. A pilot application would allow the evaluation criteria, scoring logic and criterion weights to be tested against actual substitution candidates, and refined based on the outcomes. The interactive spreadsheet tool provided in Appendix 4 was designed to support this process, with adjustable criterion weights that allow the framework to be calibrated to reflect different material contexts, module types and organizational priorities as the work progresses.

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APPENDICES

APPENDIX 1: Interview guide

The interviews were conducted in Swedish. To reflect the actual wording used during data collection, the original interview questions are presented in Swedish. English translations are provided to make the interview guides accessible within the English-language report.

Table A.1 Interview Guide for Respondent 1: Product Development and Material Decisions

Theme	**Original question in swedish**	**English translation**
Role and responsibility	Hur ser din roll ut i relation till produktutveckling och material i moduler?	How is your role related to product development and materials in the modules?
Influence on material choices	I vilken utsträckning kan ni påverka materialval, och hur sker den påverkan i praktiken?	To what extent can you influence material choices, and how does this influence take place in practice?
Previous material changes	Har ni varit med om att påverka eller ändra material i moduler?	Have you been involved in influencing or changing materials in the modules?
Evaluation factors	Vilka faktorer väger tyngst när ni utvärderar eller driver ett materialbyte?	Which factors are most important when evaluating or initiating a material change?
Trade-offs	Hur hanterar ni avvägningar mellan till exempel teknisk funktion, kostnad och hållbarhet?	How do you manage trade-offs between, for example, technical function, cost, and sustainability?
Sustainability and carbon emissions	Hur tas hållbarhet och koldioxidutsläpp hänsyn till i arbetet med material idag?	How are sustainability and carbon emissions considered in today's work with materials?
Material-specific discussion	Vi kommer även att diskutera följande material ur ett produkt- och förändringsperspektiv: Fasad skiva av pressad isolering, fibercement, glasull isolering 33 (yttervägg)	The interview will also include a discussion of the following materials from a product development and change perspective: Facade board made of compressed insulation, fibre

		cement board, glass wool insulation 33 (external wall)

Table A.2 Interview Guide for Respondent 2: Buildability and Practical Implementation

Theme	**Original question in swedish**	**English translation**
Practical function in modular building systems	Vad avgör om ett material fungerar bra i praktiken vid arbete med modulära byggsystem?	What determines whether a material works well in practice when working with modular building systems?
Technical and regulatory requirements	Vilka krav, till exempel tekniska eller regelmässiga, måste ett material uppfylla för att vara aktuellt?	Which requirements, such as technical or regulatory requirements, must a material meet in order to be considered relevant?
Experiences of problematic materials	Har du erfarenheter av material som inte fungerat bra i praktiken? Vad var i så fall utmaningen?	Do you have experience of materials that have not worked well in practice? If so, what was the challenge?
Assembly and project implementation	Hur påverkar materialval monterbarhet och genomförande i projekt?	How do material choices affect assembly and project implementation?
Feasibility of material substitution	Vad avgör om ett materialbyte är möjligt att genomföra i praktiken?	What determines whether a material substitution is possible to implement in practice?
Prioritization between factors	Hur skulle du prioritera mellan teknisk funktion, monterbarhet, kostnad, regelkrav och livslängd vid ett materialbyte?	How would you prioritize between technical function, assembly, cost, regulatory requirements and service life in a material substitution?
Material-specific discussion	Vi kommer även att diskutera följande material ur ett byggbarhetsperspektiv: Fasad skiva av pressad isolering, fibercement, glasull isolering 33 (yttervägg)	The interview will also include a discussion of the following materials from a buildability perspective: Facade board made of compressed insulation, fibre cement board, glass wool insulation 33 (external wall)

Table A.3 Interview Guide for Respondent 3: Sustainability, Climate Impact and Material Data

Theme	**Original question in swedish**	**English translation**
Role and sustainability work	Hur arbetar du med hållbarhet inom företaget, särskilt kopplat till byggprojekt eller moduler?	How do you work with sustainability within the company, particularly in relation to building projects or modules?
Materials and climate impact	I vilken utsträckning inkluderas material och klimatpåverkan i detta arbete?	To what extent are materials and climate impact included in this work?
Measurement and follow-up	Hur mäts och följs klimatpåverkan upp idag, och vilken typ av data används?	How is climate impact measured and followed up today, and what type of data is used?
Use in decision-making	Hur används denna information i praktiken, särskilt i relation till beslut?	How is this information used in practice, particularly in relation to decision-making?
Challenges	Vilka utmaningar finns i att arbeta med klimatpåverkan kopplat till material?	What challenges exist when working with climate impact related to materials?
Influence on decisions	Hur stor påverkan har hållbarhetsaspekter på faktiska beslut idag?	How much influence do sustainability aspects have on actual decisions today?
Structured way of working	Vi är även intresserade av hur ett mer strukturerat arbetssätt kring material och klimatpåverkan skulle kunna se ut.	We are also interested in what a more structured way of working with materials and climate impact could look like.

Table A.4 Interview Guide for Respondent 4 and Respondent 5: Property Management, Customer Cases and Materials in Use

Theme	**Original question in swedish**	**English translation**
Role and responsibility	Din roll i relation till arbete med moduler inom förvaltning.	Your role in relation to work with modules within property management.
Customer cases, complaints and maintenance	Erfarenheter från kundärenden, klagomål och underhåll kopplat till moduler och material.	Experiences from customer cases, complaints and maintenance related to modules and materials.
Material choices in practice	Hur materialval märks i praktiken, exempelvis genom återkommande problem eller behov av åtgärder.	How material choices become visible in practice, for example through recurring problems or the need for corrective actions.
Maintenance and service life	Hur material påverkar underhåll, livslängd och det dagliga arbetet.	How materials affect maintenance, service life and daily work.
Positive and negative experiences	Erfarenheter av situationer där materialval har fungerat bra respektive mindre bra.	Experiences of situations where material choices have worked well or less well.
Feedback to product development	Om och hur erfarenheter från förvaltning tas vidare till produktutveckling.	Whether and how experiences from property management are transferred to product development.
Customer requirements and priorities	Hur kundkrav och prioriteringar påverkar arbetet i praktiken.	How customer requirements and priorities affect the work in practice.
Improvement opportunities	Vad som skulle kunna förbättras i arbetet med material ur ett förvaltningsperspektiv.	What could be improved in the work with materials from a property management perspective.

Table A.5 Questions Answered by Respondent 6: Decision Process, Climate Data and Evaluation Criteria

Theme	**Original question in swedish**	**English translation**
Decision process for material changes	Hur går det till från att ett behov av materialförändring uppstår till att ett beslut tas, och till vilken utsträckning är detta arbete strukturerat eller situationsbaserat?	How does the process work from the point where a need for material change arises until a decision is made, and to what extent is this work structured or situation-based?
Climate data in the decision process	Hur används klimatdata, såsom EPD eller klimatdeklarationer, i beslutsprocessen och vilka utmaningar finns kopplade till detta?	How is climate data, such as EPDs or climate declarations, used in the decision process, and what challenges are associated with this?
Implementation of material substitution	Vad är det som avgör om ett materialbyte faktiskt genomförs eller inte?	What determines whether a material substitution is actually implemented or not?
Structured material selection process	Vi har, som vi berättat tidigare, jobbat med att ta fram ett mer strukturerat arbetssätt för hur materialvalsarbete skulle kunna se ut, inklusive vilka kriterier som bör ingå och hur dessa kan prioriteras. Vi vill dock veta dina tankar kring detta för att processen ska passa Expandias arbetssätt och behov.	As previously mentioned, we have worked on developing a more structured way of working with material selection, including which criteria should be included and how these can be prioritized. However, we would like to know your thoughts on this so that the process fits Expandia's way of working and needs.
Evaluation criteria	Betygsätt nedanstående sju faktorer på en skala från 1 till 5, där 5 är mycket viktig och 1 är minst viktig i utvärdering av material. Flera faktorer kan ha samma betyg: CO_2 Teknisk funktion Livslängd Kostnader Cirkularitet/återanvändning	Rate the following seven factors on a scale from 1 to 5, where 5 is very important and 1 is least important in the evaluation of materials. Several factors may receive the same rating: CO_2 Technical function Service life Costs Circularity/reuse

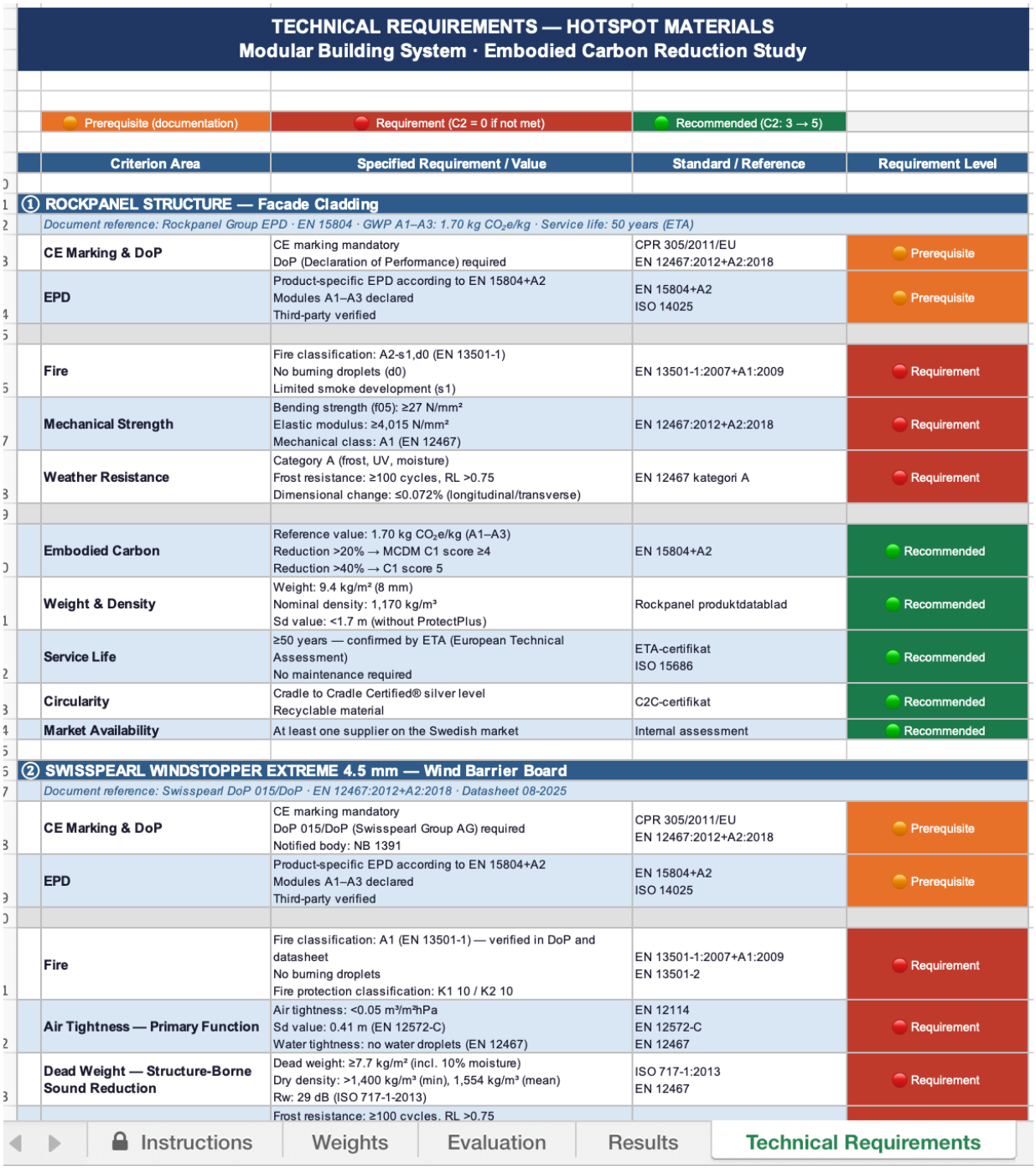
	Hanterbarhet Tillgänglighet	Manageability Availability
Additional factors	Finns det andra faktorer som är viktiga att ha med utöver de sju som vi presenterat?	Are there any other factors that are important to include in addition to the seven factors we presented?
Trade-offs between criteria	Hur resonerar ni i situationer där dessa faktorer står i konflikt, till exempel mellan kostnad och klimatpåverkan?	How do you reason in situations where these factors conflict, for example between cost and climate impact?

APPENDIX 2: Material List

Material	**Quantity (kg)**	**Emission factor (EPD kg CO_2e/kg)**
Facade board made of compressed insulation	3130,268352	1,7
Fibre cement board	6876,3816	0,65
Glass wool insulation 33, external wall	4479,529	0,89
PVC window	940,8	2,63
Particle board, external wall	4878,72	0,488
PVC roof membrane	316,8	7,05
Plasterboard, internal wall	7092,442	0,284
Plasterboard, external wall	6936,16	0,284
Particle board, internal wall	2958,52704	0,488
Plywood	2935,6074	0,488

Studs, building envelope	13130,37945	0,0919
Glulam	8354,7753	0,12
Gypsum fibreboard, internal wall	1863	0,495
Glass wool insulation 33, internal wall	803,6938813	1,11
Nailing batten, behind facade	479,1024	1,7
Gypsum board, wind barrier	1416,71628	0,333
Aluminium-glass, external door	96,6	4,88
Glass wool insulation 36	332,64	1,11
Sleepers	3054,4	0,08
Solid laminate, internal door	535	0,388
Studs, internal wall	1148,273775	0,0919
Wood, internal staircase	588	0,0919
Glazed laminate, internal door	41	0,388

APPENDIX 3: Technical Requirements per Hotspot Material



TECHNICAL REQUIREMENTS — HOTSPOT MATERIALS
Modular Building System · Embodied Carbon Reduction Study

Prerequisite (documentation) | Requirement (C2 = 0 if not met) | Recommended (C2: 3 → 5)

Criterion Area	Specified Requirement / Value	Standard / Reference	Requirement Level
① ROCKPANEL STRUCTURE — Facade Cladding			
Document reference: Rockpanel Group EPD · EN 15804 · GWP A1–A3: 1.70 kg CO_2e/kg · Service life: 50 years (ETA)			
CE Marking & DoP	CE marking mandatory DoP (Declaration of Performance) required	CPR 305/2011/EU EN 12467:2012+A2:2018	Prerequisite
EPD	Product-specific EPD according to EN 15804+A2 Modules A1–A3 declared Third-party verified	EN 15804+A2 ISO 14025	Prerequisite
Fire	Fire classification: A2-s1,d0 (EN 13501-1) No burning droplets (d0) Limited smoke development (s1)	EN 13501-1:2007+A1:2009	Requirement
Mechanical Strength	Bending strength (f05): ≥27 N/mm² Elastic modulus: ≥4,015 N/mm² Mechanical class: A1 (EN 12467)	EN 12467:2012+A2:2018	Requirement
Weather Resistance	Category A (frost, UV, moisture) Frost resistance: ≥100 cycles, RL >0.75 Dimensional change: ≤0.072% (longitudinal/transverse)	EN 12467 kategori A	Requirement
Embodied Carbon	Reference value: 1.70 kg CO_2e/kg (A1–A3) Reduction >20% → MCDM C1 score ≥4 Reduction >40% → C1 score 5	EN 15804+A2	Recommended
Weight & Density	Weight: 9.4 kg/m² (8 mm) Nominal density: 1,170 kg/m³ Sd value: <1.7 m (without ProtectPlus)	Rockpanel produktdatablad	Recommended
Service Life	≥50 years — confirmed by ETA (European Technical Assessment) No maintenance required	ETA-certifikat ISO 15686	Recommended
Circularity	Cradle to Cradle Certified® silver level Recyclable material	C2C-certifikat	Recommended
Market Availability	At least one supplier on the Swedish market	Internal assessment	Recommended
② SWISSPEARL WINDSTOPPER EXTREME 4.5 mm — Wind Barrier Board			
Document reference: Swisspearl DoP 015/DoP · EN 12467:2012+A2:2018 · Datasheet 08-2025			
CE Marking & DoP	CE marking mandatory DoP 015/DoP (Swisspearl Group AG) required Notified body: NB 1391	CPR 305/2011/EU EN 12467:2012+A2:2018	Prerequisite
EPD	Product-specific EPD according to EN 15804+A2 Modules A1–A3 declared Third-party verified	EN 15804+A2 ISO 14025	Prerequisite
Fire	Fire classification: A1 (EN 13501-1) — verified in DoP and datasheet No burning droplets Fire protection classification: K1 10 / K2 10	EN 13501-1:2007+A1:2009 EN 13501-2	Requirement
Air Tightness — Primary Function	Air tightness: <0.05 m³/m²hPa Sd value: 0.41 m (EN 12572-C) Water tightness: no water droplets (EN 12467)	EN 12114 EN 12572-C EN 12467	Requirement
Dead Weight — Structure-Borne Sound Reduction	Dead weight: ≥7.7 kg/m² (incl. 10% moisture) Dry density: >1,400 kg/m³ (min), 1,554 kg/m³ (mean) Rw: 29 dB (ISO 717-1-2013)	ISO 717-1:2013 EN 12467	Requirement
	Frost resistance: ≥100 cycles, RL >0.75		

Instructions | Weights | Evaluation | Results | Technical Requirements

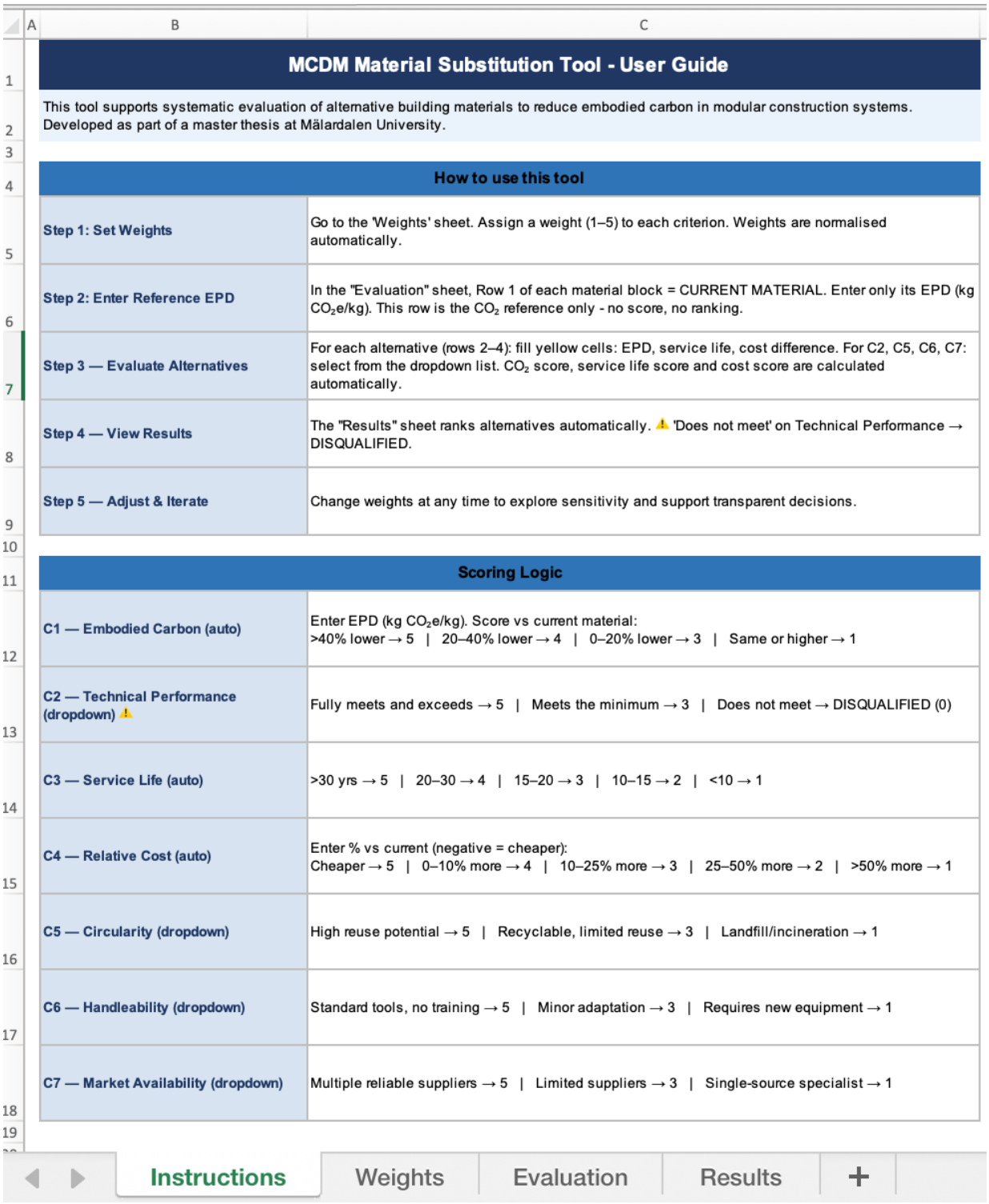
	A	B	C	D	E
34		**Weather Resistance**	Frost resistance: ≥100 cycles, RL >0.75 Moisture movement: 0.55 mm/m (30/90% RH) Water absorption: 19.9% (24h)	EN 12467	Requirement
35					
36		**Embodied Carbon**	Reference value: 0.65 kg CO_2e/kg (A1–A3) Reduction >20% → MCDM C1 score ≥4	EN 15804+A2	Recommended
37		**Geometry & Format**	Thickness: 4.5 mm Standard dimensions: 900×2700, 900×3000, 1200×2700, 1200×3000 mm Thickness tolerance: ±0.6 mm	EN 12467 nivå 1 Swisspearl datablad	Recommended
38		**Mechanical Properties**	Bending strength longitudinal: 17.5 MPa, transverse: 22.1 MPa E-modulus longitudinal: 16.9 GPa, transverse: 16.1 GPa Impact resistance: Approved category IV	EN 12467 EAD 090062-00-0404	Recommended
39		**Service Life**	≥20 years (climate envelope placement)	ISO 15686	Recommended
40					
41		**③ ISOVER UNI 33 — Thermal Insulation (glass wool)**			
42		*Document reference: EPD-IES-0026646:001 · EN 15804+A2:2019/AC:2021 · Valid until 2031-01-13 · Billesholm, Sweden*			
43		**CE Marking & DoP**	CE marking mandatory (EN 13162) DoP required	CPR 305/2011/EU EN 13162:2012+A1:2015	Prerequisite
44		**EPD**	Product-specific EPD according to EN 15804+A2 Modules A1–A3 declared Third-party verified Nordic geographical coverage	EN 15804+A2:2019/AC:2021 ISO 14025 PCR 2019:14 v2.0.1	Prerequisite
45					
46		**Fire**	Fire classification: A1 (EN 13501-1) — verified in EPD Does not melt or fall away in fire	EN 13501-1:2007+A1:2009 EN 13162	Requirement
47		**Thermal Performance — Lambda Value**	λ value: 0.033 W/(m·K) — verified in EPD R value: 1 K·m²/W at 33 mm (comparison basis) U value of the wall must not deteriorate	EN 12667 / EN 12939 BBR 9:2 (U-krav Sverige)	Requirement
48		**Construction Compatibility**	Standard thicknesses compatible with module stud depth Available thicknesses: 45, 70, 95, 120, 145, 170, 195, 220 mm Easy cutting with standard tools	Isover EPD — produktbeskrivning	Requirement
49					
50		**Embodied Carbon**	Reference value: ≈0.89 kg CO_2e/kg (A1–A3) Source: EPD-IES-0026646:001, GWP-total 4.88E-01 kg CO_2e per DU (0.78 kg) Reduction >20% → MCDM C1 score ≥4	EN 15804+A2 EPD-IES-0026646:001	Recommended
51		**Density & Weight**	Density: 23.5 kg/m³ Mass per DU: 0.78 kg (33 mm reference thickness)	EPD-IES-0026646:001 sida 3	Recommended
52		**Acoustic Performance**	Airborne sound absorption must not deteriorate vs. reference product Absorption coefficient (αw) to be documented	EN ISO 354 EN ISO 717-1	Recommended
53		**Service Life**	RSL: 50 years — declared in EPD No maintenance required during service life Correct installation assumed	EPD-IES-0026646:001 sida 4–5 ISO 15686	Recommended
54		**Circularity**	Recycled glass content: 50–60% (post-consumer) Recyclable material	EPD-IES-0026646:001 innehållsdeklaration	Recommended
55		**Market Availability**	Several established suppliers on the Swedish market E.g.: ISOVER (Billesholm), Rockwool, Knauf, Paroc	Internal assessment	Recommended
56					
57		**GENERAL REQUIREMENTS — Applicable to all three hotspot materials**			
58		**CE Marking & DoP**	Mandatory for all three materials (CPR 305/2011/EU)		
59		**EPD-standard**	EN 15804+A1 or +A2 — A1–A3 modules declared. Same PCR as reference product for comparability.		
60		**MCDM C2 Link**	The requirement level is directly linked to C2. Not met → C2 = 0 → alternative is disqualified.		
61		**Nordic Market**	Product available for delivery to Sweden.		
62		**EPD Comparability**	For EPDs to be comparable, the same PCR, same system boundaries and same declared unit are required.		
63		**Document Quality**	Product-specific, third-party verified EPD provides stronger basis for C1 score 5 compared to industry EPD.		
64					

Sources: Personal communication with industry supervisor (case study company). Swisspearl DoP 015/DoP (CE_2013_015DoP). Swisspearl Datasheet Windstopper Extreme/Connect 08-2025. Isover EPD EPD-IES-0026646:001

Instructions | Weights | Evaluation | Results | **Technical Requirements**

APPENDIX 4: MCDM Interactive Evaluation Tool

Figure A.1. Instructions sheet



MCDM Material Substitution Tool - User Guide

This tool supports systematic evaluation of alternative building materials to reduce embodied carbon in modular construction systems.
Developed as part of a master thesis at Mälardalen University.

How to use this tool

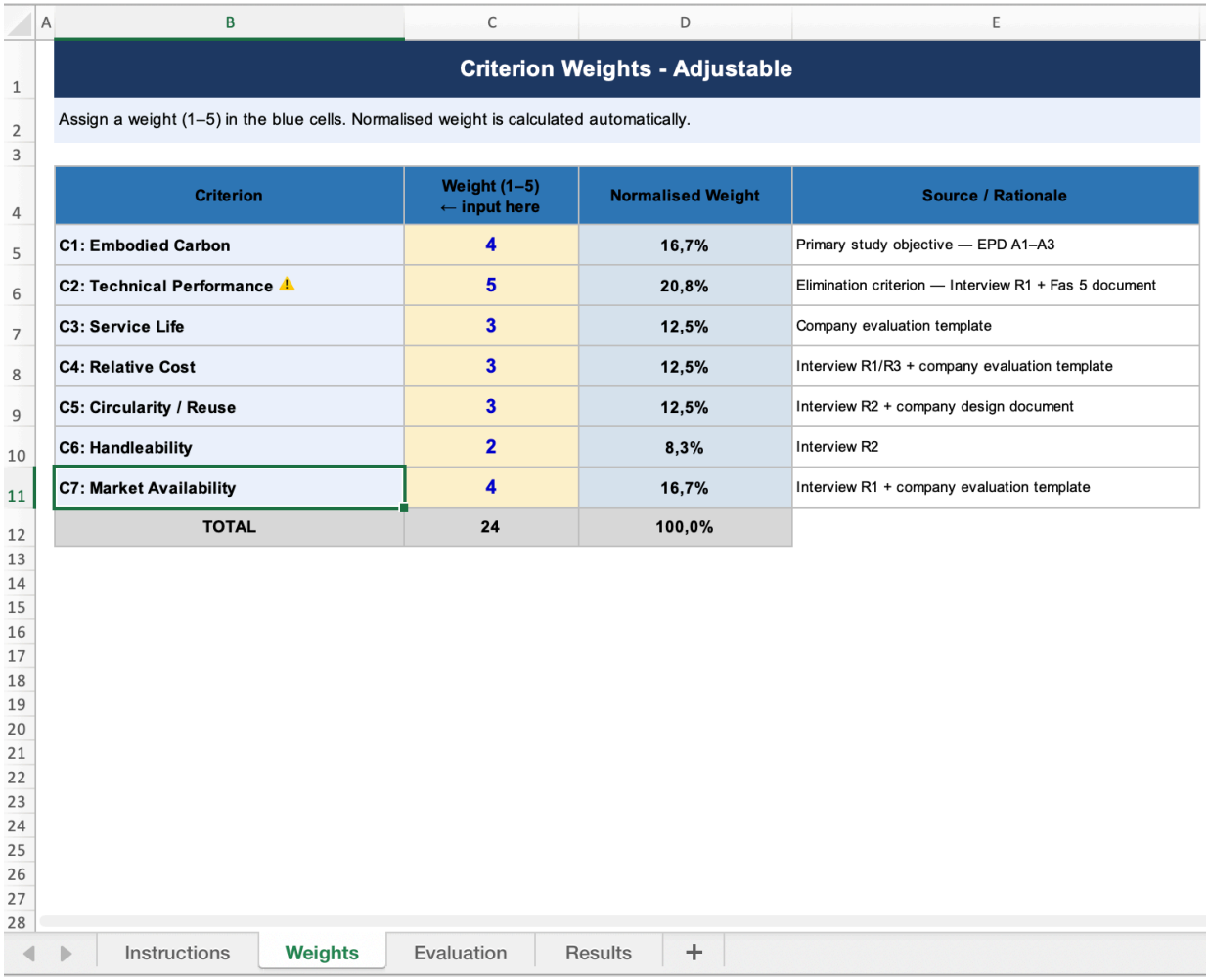
Step 1: Set Weights	Go to the 'Weights' sheet. Assign a weight (1–5) to each criterion. Weights are normalised automatically.
Step 2: Enter Reference EPD	In the "Evaluation" sheet, Row 1 of each material block = CURRENT MATERIAL. Enter only its EPD (kg CO_2e/kg). This row is the CO_2 reference only - no score, no ranking.
Step 3 — Evaluate Alternatives	For each alternative (rows 2–4): fill yellow cells: EPD, service life, cost difference. For C2, C5, C6, C7: select from the dropdown list. CO_2 score, service life score and cost score are calculated automatically.
Step 4 — View Results	The "Results" sheet ranks alternatives automatically. ⚠ 'Does not meet' on Technical Performance → DISQUALIFIED.
Step 5 — Adjust & Iterate	Change weights at any time to explore sensitivity and support transparent decisions.

Scoring Logic

C1 — Embodied Carbon (auto)	Enter EPD (kg CO_2e/kg). Score vs current material: >40% lower → 5 | 20–40% lower → 4 | 0–20% lower → 3 | Same or higher → 1
C2 — Technical Performance (dropdown) ⚠	Fully meets and exceeds → 5 | Meets the minimum → 3 | Does not meet → DISQUALIFIED (0)
C3 — Service Life (auto)	>30 yrs → 5 | 20–30 → 4 | 15–20 → 3 | 10–15 → 2 | <10 → 1
C4 — Relative Cost (auto)	Enter % vs current (negative = cheaper): Cheaper → 5 | 0–10% more → 4 | 10–25% more → 3 | 25–50% more → 2 | >50% more → 1
C5 — Circularity (dropdown)	High reuse potential → 5 | Recyclable, limited reuse → 3 | Landfill/incineration → 1
C6 — Handleability (dropdown)	Standard tools, no training → 5 | Minor adaptation → 3 | Requires new equipment → 1
C7 — Market Availability (dropdown)	Multiple reliable suppliers → 5 | Limited suppliers → 3 | Single-source specialist → 1

Instructions | Weights | Evaluation | Results | +

Figure A.2. Weights sheet



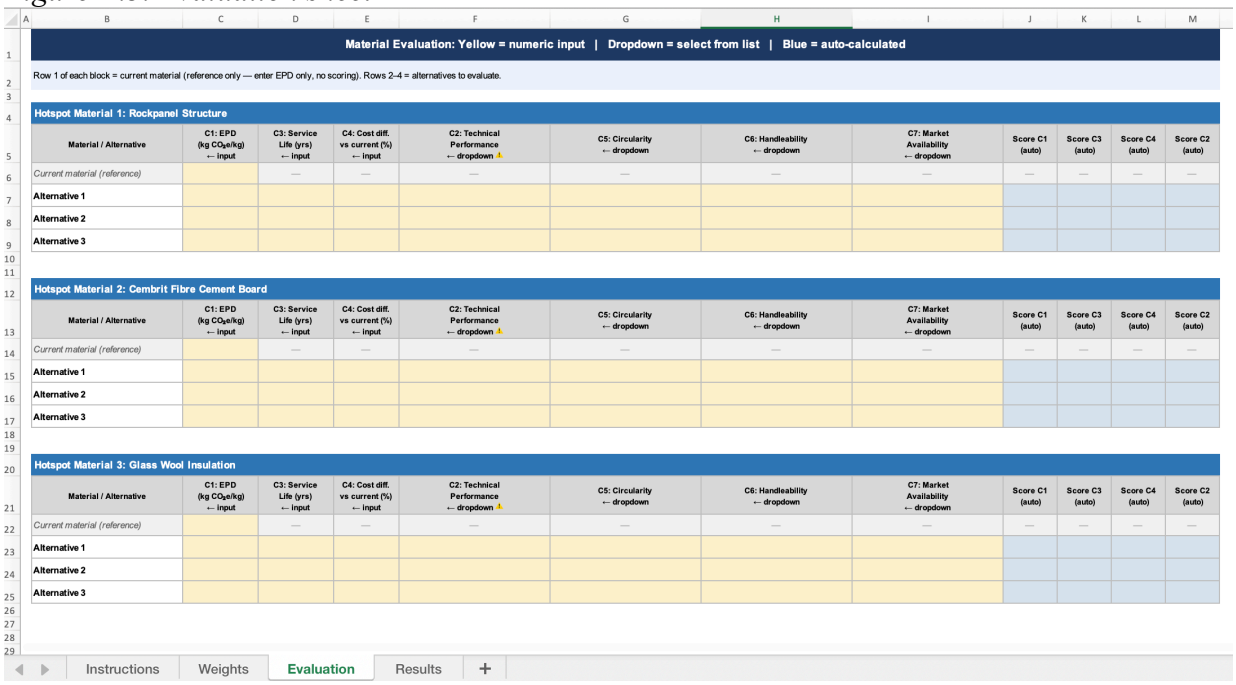
Criterion Weights - Adjustable

Assign a weight (1–5) in the blue cells. Normalised weight is calculated automatically.

Criterion	Weight (1–5) ← input here	Normalised Weight	Source / Rationale
C1: Embodied Carbon	4	16,7%	Primary study objective — EPD A1–A3
C2: Technical Performance ⚠	5	20,8%	Elimination criterion — Interview R1 + Fas 5 document
C3: Service Life	3	12,5%	Company evaluation template
C4: Relative Cost	3	12,5%	Interview R1/R3 + company evaluation template
C5: Circularity / Reuse	3	12,5%	Interview R2 + company design document
C6: Handleability	2	8,3%	Interview R2
C7: Market Availability	4	16,7%	Interview R1 + company evaluation template
TOTAL	24	100,0%	

Instructions | Weights | Evaluation | Results | +

Figure A.3. Evaluation sheet



Material Evaluation: Yellow = numeric input | Dropdown = select from list | Blue = auto-calculated

Row 1 of each block = current material (reference only — enter EPD only, no scoring). Rows 2–4 = alternatives to evaluate.

Hotspot Material 1: Rockpanel Structure

Material / Alternative	C1: EPD (kg CO_2e/kg) ← input	C3: Service Life (yrs) ← input	C4: Cost diff. vs current (%) ← input	C2: Technical Performance ← dropdown ⚠	C5: Circularity ← dropdown	C6: Handleability ← dropdown	C7: Market Availability ← dropdown	Score C1 (auto)	Score C3 (auto)	Score C4 (auto)	Score C2 (auto)
Current material (reference)		—	—	—	—	—	—	—	—	—	—
Alternative 1											
Alternative 2											
Alternative 3											

Hotspot Material 2: Cembrit Fibre Cement Board

Material / Alternative	C1: EPD (kg CO_2e/kg) ← input	C3: Service Life (yrs) ← input	C4: Cost diff. vs current (%) ← input	C2: Technical Performance ← dropdown ⚠	C5: Circularity ← dropdown	C6: Handleability ← dropdown	C7: Market Availability ← dropdown	Score C1 (auto)	Score C3 (auto)	Score C4 (auto)	Score C2 (auto)
Current material (reference)		—	—	—	—	—	—	—	—	—	—
Alternative 1											
Alternative 2											
Alternative 3											

Hotspot Material 3: Glass Wool Insulation

Material / Alternative	C1: EPD (kg CO_2e/kg) ← input	C3: Service Life (yrs) ← input	C4: Cost diff. vs current (%) ← input	C2: Technical Performance ← dropdown ⚠	C5: Circularity ← dropdown	C6: Handleability ← dropdown	C7: Market Availability ← dropdown	Score C1 (auto)	Score C3 (auto)	Score C4 (auto)	Score C2 (auto)
Current material (reference)		—	—	—	—	—	—	—	—	—	—
Alternative 1											
Alternative 2											
Alternative 3											

Instructions | Weights | Evaluation | Results | +

Figure A.4. Results sheet

MCDM Results: Weighted Score Ranking

Scores calculated automatically. Higher = more suitable. DISQUALIFIED = Technical Performance 'Does not meet'.

Hotspot Material 1: Rockpanel Structure

Alternative	Weighted Score (0–5)	CO_2 Reduction vs Current	Rank
Alternative 1	Incomplete	—	—
Alternative 2	Incomplete	—	—
Alternative 3	Incomplete	—	—

Hotspot Material 2: Cembrit Fibre Cement Board

Alternative	Weighted Score (0–5)	CO_2 Reduction vs Current	Rank
Alternative 1	Incomplete	—	—
Alternative 2	Incomplete	—	—
Alternative 3	Incomplete	—	—

Hotspot Material 3: Glass Wool Insulation

Alternative	Weighted Score (0–5)	CO_2 Reduction vs Current	Rank
Alternative 1	Incomplete	—	—
Alternative 2	Incomplete	—	—
Alternative 3	Incomplete	—	—

Instructions | Weights | Evaluation | Results | +